Saviours AI

ICSE CLASS 10 · 2027 SYLLABUS

SIDE KICK

Biology

Sixteen chapters written fresh for the 2027 syllabus: notes, diagrams, mind maps, HOT questions and a practical lab companion.



EVERY TOPPER HAS A SIDEKICK

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Hey. I'm your Sidekick.

I live inside this book. Wherever things get tricky, I pop up with a tip, a trap to dodge, or a question the board loves to ask. Read the chapter, then try my HOT questions. Deal?



Aligned to the official ICSE 2027 syllabus. If it's in the syllabus, it's in here.

SIDEKICK TIP
Shortcuts, mnemonics and exam hacks.

FUN FACT
Things that make the concept stick.

REAL LIFE
Where this shows up outside the exam hall.

KEY POINT
Definitions and laws you must write exactly.

HOT QUESTION
Higher-order thinking. Try before you peek.

✦ NEW
Added in this edition to match the 2027 syllabus.

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01

CHAPTER · BIOLOGY

Cell Cycle and Cell Division

Your body began as a single cell and now has trillions. This chapter follows the life of one cell: how it prepares, how it splits in two, and how a different kind of division makes the cells that carry life to the next generation.



IN THIS CHAPTER

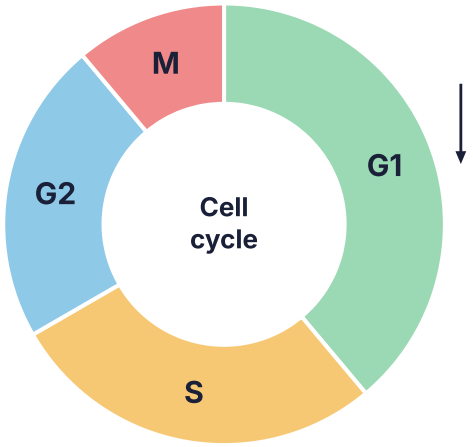
- Cell cycle
- Interphase and M phase
- Mitosis stages
- Meiosis in brief
- Crossing over
- Mitosis vs meiosis

The Cell Cycle

A cell does not simply split when it feels like it. It follows a fixed sequence of events, from the moment it is born to the moment it divides into daughter cells. This ordered sequence is the **cell cycle**.

KEY POINT

Cell cycle: the series of events through which a cell grows, copies its DNA and divides to produce two daughter cells.



G1 (gap 1)
Cell grows, makes proteins and organelles

S (synthesis)
DNA is copied; each chromosome gets a twin

G2 (gap 2)
Energy stored, spindle proteins made

M (mitotic phase)
Mitosis, then cytokinesis

G1 + S + G2 = interphase (not to scale)

FIG 1.1 THE CELL CYCLE

The cycle has two big parts. **Interphase** is the long preparation period in which the cell looks calm under the microscope but is busy inside. The **mitotic phase (M phase)** is the short, dramatic part when the nucleus divides (**karyokinesis**) and the cytoplasm divides (**cytokinesis**). Most of a cell's life is spent in interphase.

PHASE	ALSO CALLED	WHAT HAPPENS
G_1	First gap phase	The new cell grows in size, makes proteins and new organelles, and carries out its normal job.
S	Synthesis phase	The DNA is copied (replicated). Every chromosome becomes a pair of identical chromatids, so the DNA content doubles.
G_2	Second gap phase	The cell keeps growing, stores energy and makes the proteins needed to build the spindle. It checks that the copied DNA is complete.
M	Mitotic phase	Mitosis (division of the nucleus) followed by cytokinesis (division of the cytoplasm).

SIDEKICK TIP

G1, S, G2, M = 'Grow, Set to copy, Get ready, Move apart'. The letter S is the only one that is about DNA; the two G's are about growth.

Mitosis

Mitosis is an **equational division**: one parent cell produces two daughter cells with exactly the same number and kind of chromosomes as the parent. It takes place in the body (somatic) cells. The human body cell has 46 chromosomes (2n = 46), and each daughter cell also gets 46.

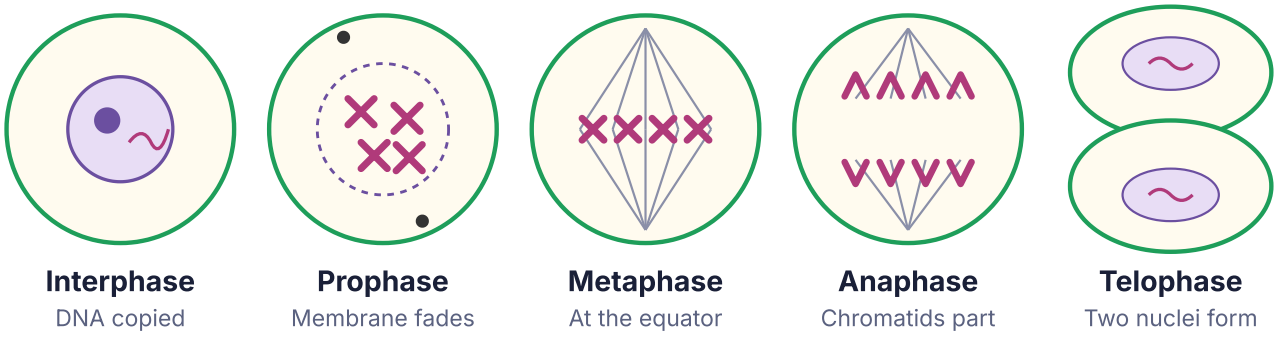


FIG 1.2 STAGES OF MITOSIS IN AN ANIMAL CELL (SIMPLIFIED, 4 CHROMOSOMES)

1. Prophase
The chromatin threads coil and shorten into visible chromosomes, each made of two chromatids joined at the centromere. The nucleolus and nuclear membrane disappear. In animal cells the two centrioles move to opposite poles and the spindle starts to form.

2. Metaphase
The chromosomes line up along the middle of the cell, the equatorial plate. Each is attached by its centromere to spindle fibres from both poles. This is the best stage for counting chromosomes.

3. Anaphase
The centromere of every chromosome splits. The two chromatids, now called daughter chromosomes, are pulled to opposite poles by the shortening spindle fibres.

4. Telophase
Chromosomes reach the poles and uncoil into chromatin. A nuclear membrane forms around each set and the nucleolus returns. The spindle disappears.

Cytokinesis: dividing the cytoplasm

- **Animal cell:** the cell membrane pinches inward in the middle, forming a **cleavage furrow** that deepens until the cell is cut in two.
- **Plant cell:** the rigid cell wall stops pinching. Instead a **cell plate** forms in the middle from the centre outward and becomes the new wall between the two cells.
- Higher plant cells have no centrioles, so their spindle forms without them.

BOARD ALERT
Do not write that chromosomes 'split' in prophase. In prophase they are already double (two chromatids). They separate only in **anaphase**, when the centromere divides.

Why mitosis matters

- **Growth:** a fertilised egg becomes an adult through repeated mitosis.
- **Repair and replacement:** healing of wounds, replacement of worn skin cells and blood cells.
- **Asexual reproduction:** yeast budding, Amoeba dividing, vegetative propagation in plants.

- **Stable chromosome number:** every cell of the body gets a full, identical set of genetic instructions.

Meiosis: the Reduction Division

Meiosis happens only in the reproductive organs (testes and ovaries in humans) when **gametes** are formed. One diploid cell (2n) divides twice in quick succession to give **four haploid cells (n)**, each with half the chromosome number. In humans, a cell with 46 chromosomes gives sperm or eggs with 23. Without this halving, the chromosome number would double in every generation. The syllabus asks for the idea only, not the stages.

Homologous chromosomes and crossing over

- **Homologous chromosomes** are the two chromosomes of a pair, one inherited from each parent. They are the same size and shape and carry genes for the same traits at the same positions, although they may carry different versions of those genes.
- Early in meiosis, homologous chromosomes come together and lie side by side. The point where they touch is called a **chiasma**.
- **Crossing over** is the exchange of corresponding segments between the non-sister chromatids of homologous chromosomes.
- After crossing over the chromatids carry new combinations of genes. The gametes are therefore different from each other and from the parent, producing **variation**.



FUN FACT

Because of crossing over and the random way chromosomes are shared out, one person can in theory make over eight million genetically different eggs or sperm, even before crossing over is counted.

Mitosis and Meiosis Compared

FEATURE	MITOSIS	MEIOSIS
Where	Body (somatic) cells; growing tissues	Reproductive organs, to form gametes
Number of divisions	One	Two, in succession
Daughter cells	Two	Four
Chromosome number	Same as parent (2n to 2n)	Halved (2n to n)
Pairing of homologous chromosomes	Does not occur	Occurs, with crossing over
Genetic make-up of daughters	Identical to parent and to each other	Different from parent and from each other
Purpose	Growth, repair, asexual reproduction	Gamete formation; keeps chromosome number constant; variation
Type of division	Equational	Reductional (first division)

REAL LIFE

Cancer is mitosis that has escaped control. Cells ignore the signals that tell them to stop dividing and pile up as a tumour. Treatments such as chemotherapy target rapidly dividing cells, which is why they also affect hair and the lining of the gut.

HOT QUESTION

A root-tip cell of onion has 16 chromosomes. How many chromosomes and how many chromatids will there be in this cell at (a) metaphase and (b) anaphase (counting each pole separately), and what will each of the two daughter cells contain?

HOT QUESTION

A student claims that crossing over also occurs in mitosis and is the reason skin cells are all alike. Explain what is wrong with this statement.

Mind Map Recap

BRANCH	WHAT TO REMEMBER
Cell cycle	Interphase (G1, S, G2) + M phase (mitosis + cytokinesis). Longest part: interphase. DNA copied in S.
Mitosis	Prophase, metaphase, anaphase, telophase. 2n → two cells of 2n. Growth, repair, asexual reproduction.
Cytokinesis	Animal: cleavage furrow. Plant: cell plate.
Meiosis	Two divisions, four haploid cells, in gonads. Halves chromosome number.
Variation	Homologous chromosomes pair; crossing over between non-sister chromatids at chiasma.

Practice Questions

#	QUESTION	ANSWER
1	Name the three sub-phases of interphase and state the main event of each.	G1: growth of the cell; S: DNA replication; G2: preparation for division, synthesis of spindle proteins.
2	Why is metaphase the best stage to count chromosomes?	The chromosomes are fully condensed and arranged in one plane at the equator, so they are clearly visible and separate.
3	What is the difference between a chromosome and a chromatid?	A chromatid is one of the two identical strands of a duplicated chromosome. Two sister chromatids joined at a centromere make one chromosome during prophase and metaphase.
4	In which phase does the centromere divide?	Anaphase.
5	How does cytokinesis in a plant cell differ from that in an animal cell?	Plant: a cell plate forms from the centre outward. Animal: a cleavage furrow pinches the cell from outside inward.
6	Define crossing over and state its biological significance.	Exchange of segments between non-sister chromatids of homologous chromosomes during meiosis. It produces new gene combinations, so offspring show variation.
7	Why must gametes be haploid?	At fertilisation two gametes fuse. If each were diploid, the zygote would have double the number of chromosomes and the number would keep doubling in each generation.
8	State any three differences between mitosis and meiosis.	Mitosis has one division, gives two identical diploid cells and occurs in body cells; meiosis has two divisions, gives four haploid genetically different cells and occurs in gonads.

02

CHAPTER · BIOLOGY

Structure of Chromosome

Inside every nucleus is a library of instructions written in DNA. This chapter opens the library: how DNA is packed into chromosomes, what a gene is, and why the double helix is such a clever design.



IN THIS CHAPTER

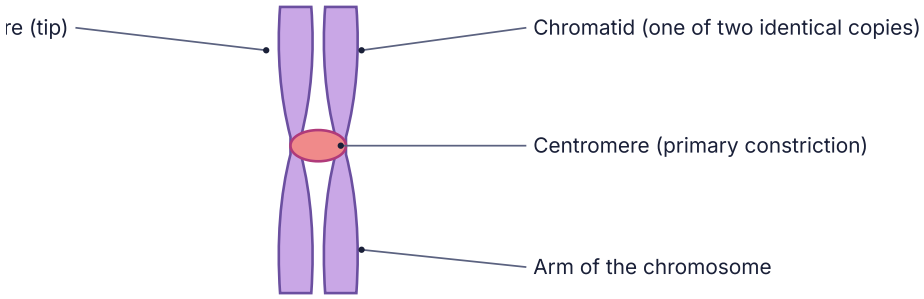
- Chromatin
- Chromosome
- Chromatid and centromere
- Genes
- DNA structure
- DNA replication

From Chromatin to Chromosome

The nucleus of a non-dividing cell contains a tangled mass of very fine threads called **chromatin**. Chromatin is made of **DNA wrapped around histone proteins**. When a cell is about to divide, the chromatin coils and shortens, becoming thick enough to be seen under a microscope as **chromosomes**. After division the chromosomes uncoil back into chromatin.

KEY POINT

Chromatin is the thin, thread-like, uncoiled form of genetic material in a resting nucleus. **Chromosome** is its condensed, rod-like form during cell division. They are the same material in two states.



A duplicated chromosome as seen in prophase or metaphase

FIG 2.1 STRUCTURE OF A CHROMOSOME

TERM	MEANING
Chromatid	One of the two identical halves of a duplicated chromosome. Two **sister chromatids** are made when DNA is copied in the S phase.
Centromere	The narrow region (primary constriction) holding the two chromatids together. Spindle fibres attach to it, and it divides at the start of anaphase.
Arms	The parts of the chromatids on either side of the centromere.
Telomere	The protective end cap of a chromosome.
Gene	A segment of DNA at a fixed position on a chromosome that carries the instruction for one trait (usually for making one protein).

Chromosomes are made mainly of **DNA and proteins**. Every species has a characteristic chromosome number. Humans have **46 chromosomes arranged as 23 pairs**. Of these, 22 pairs are **autosomes** (control body traits) and one pair is the **sex chromosomes** (XX in females, XY in males). A gamete carries 23 chromosomes.

ORGANISM	BODY CELL CHROMOSOME NUMBER (2N)
Human	46
Chimpanzee	48
Dog	78
Onion	16
Garden pea	14
Fruit fly	8

FUN FACT

Chromosome number says nothing about how complex an organism is. A dog has more chromosomes than a human, and a fern can have hundreds.

DNA: The Molecule of Heredity

DNA (deoxyribonucleic acid) is the genetic material of nearly all living things. Its structure, a **double helix**, was worked out in 1953 by James Watson and Francis Crick, using X-ray pictures made by Rosalind Franklin and Maurice Wilkins.

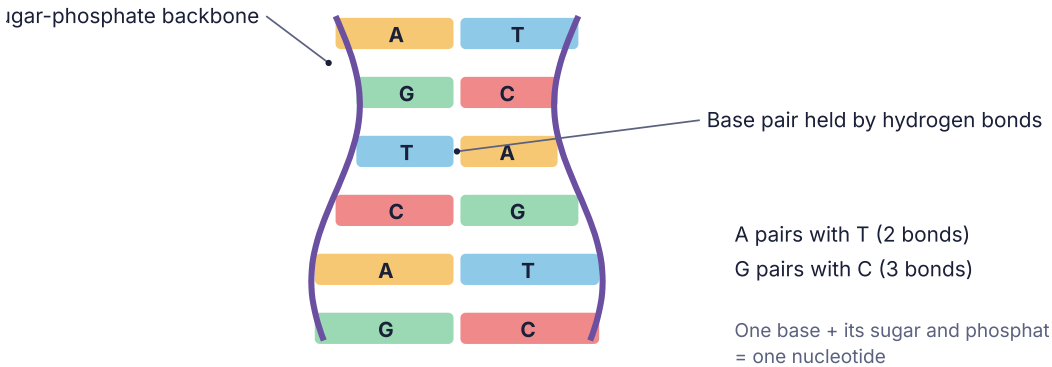


FIG 2.2 A STRETCH OF THE DNA DOUBLE HELIX (SIMPLIFIED LADDER VIEW)

- DNA is a long molecule made of repeating units called **nucleotides**. Each nucleotide has a deoxyribose sugar, a phosphate group and one nitrogen-containing **base**.
- There are four bases: **adenine (A), thymine (T), guanine (G) and cytosine (C)**.
- Two strands twist around each other like a spiral ladder. The sugar-phosphate chains are the sides; the pairs of bases are the rungs.
- Bases pair by **complementary base pairing**: A with T (two hydrogen bonds) and G with C (three hydrogen bonds). If one strand reads ATGC, the other reads TACG.
- The **sequence of bases** along the strand is the genetic code. Different orders spell out different genes.

SIDEKICK TIP

'Apple in the Tree, Car in the Garage' gives you the pairs: A-T and C-G.

DNA replication

Before a cell divides, its DNA must be copied. The two strands unzip, and each strand acts as a template on which a new complementary strand is built. The result is two DNA molecules, each with one old and one new strand. This is why daughter cells receive an exact copy of the genetic message. Replication takes place in the **S phase** of interphase.

Genes and What They Do

- A gene is a stretch of DNA. A chromosome holds hundreds to thousands of genes in a linear order.
- A gene carries the code for a protein. Proteins (enzymes, pigments, hormones, structural proteins) decide the visible traits, such as eye colour or blood group.
- Humans have roughly 20,000 to 25,000 genes. Different forms of the same gene are called **alleles**.
- A change in the base sequence is a **mutation**. It may alter the protein made, and the trait with it.

FEATURE	DNA	CHROMOSOME
What is it?	A molecule	A structure made of DNA and protein
Visible under a light microscope?	No	Yes, during cell division
Relationship	One chromosome has one long DNA molecule (two after S phase)	Contains many genes along its length

REAL LIFE

DNA fingerprinting works because, apart from identical twins, no two people have exactly the same DNA sequence in certain repeating regions. It is used in crime investigations and to settle parentage.

BOARD ALERT

Do not call the nucleus 'the DNA' or the gene 'the chromosome'. The order of size is: base < nucleotide < gene < chromosome < genome.

HOT QUESTION

A segment of one strand of DNA reads A-A-G-C-T-T-G. Write the sequence of the complementary strand. If this segment has 7 base pairs, how many hydrogen bonds hold the two strands together here?

Mind Map Recap

BRANCH	WHAT TO REMEMBER
Chromatin	Thin uncoiled DNA + histone threads in a resting nucleus
Chromosome	Condensed rod form in division; two sister chromatids joined at centromere
Human number	46 = 22 pairs autosomes + 1 pair sex chromosomes
DNA	Double helix; sugar-phosphate backbone; bases A-T, G-C
Gene	Segment of DNA coding for a protein/trait; alleles = alternative forms
Replication	Semi-conservative; S phase; template strands

Practice Questions

#	QUESTION	ANSWER
1	Differentiate between chromatin and chromosome.	Chromatin is the thin, extended thread form seen in a resting nucleus; a chromosome is the condensed, thick rod form seen in a dividing cell.
2	What is a centromere? State its role.	The constricted region joining two chromatids. Spindle fibres attach to it and it divides in anaphase so chromatids can move apart.
3	Name the four bases of DNA and state the base-pairing rule.	Adenine, thymine, guanine, cytosine. A pairs with T and G pairs with C.
4	Who proposed the double helix model of DNA and in which year?	Watson and Crick, in 1953.
5	Why is the DNA of two daughter cells identical?	Because DNA is replicated in S phase using each old strand as a template, giving two identical copies before the cell divides.
6	How many autosomes and sex chromosomes are present in a human body cell?	44 autosomes (22 pairs) and 2 sex chromosomes (one pair).
7	Define a gene.	A segment of DNA at a particular position on a chromosome that carries the instruction for a specific trait, usually by coding for a protein.
8	How many chromosomes will a human sperm contain, and why?	23, because meiosis halves the number of chromosomes from 46.

03

CHAPTER · BIOLOGY

Genetics

Why do you have your mother's nose and your father's height, and why can a trait skip a generation? Genetics gives the rules. Gregor Mendel found them with a garden of pea plants, long before anyone had seen a gene.



IN THIS CHAPTER

- Genetic terms
- Mendel's three laws
- Monohybrid cross
- Dihybrid cross
- Sex determination
- Haemophilia and colour blindness

Vocabulary of Genetics

Genetics is the study of heredity (the passing of traits from parents to offspring) and variation (the differences between individuals). Get these terms right first, because every exam answer is built from them.

TERM	MEANING	EXAMPLE
Gene	Unit of heredity; a segment of DNA controlling a trait	Gene for height in pea
Allele	One of the alternative forms of a gene	T (tall) and t (dwarf)
Dominant	Allele that expresses itself even when present with another allele	T
Recessive	Allele that is expressed only when two copies are present	t
Homozygous	Two identical alleles for a trait (pure)	TT or tt
Heterozygous	Two different alleles for a trait (hybrid)	Tt
Genotype	The genetic make-up for a trait	TT, Tt, tt
Phenotype	The visible or expressed trait	Tall or dwarf
Variation	Differences between individuals of a species	Height, skin colour
Mutation	A sudden, permanent change in a gene or chromosome that can be inherited	Albinism



KEY POINT

A **dominant** allele is written with a capital letter and a **recessive** with the small letter of the same alphabet. A tall plant can be TT or Tt, but a dwarf plant can only be tt.

Mendel's Experiments

Gregor Johann Mendel, an Austrian monk, bred garden pea plants (*Pisum sativum*) in the 1850s and 1860s. Peas were an excellent choice: they grow quickly, have clear contrasting traits, self-pollinate, and can be cross-pollinated by hand. Mendel studied **seven traits**, one at a time at first, and counted the offspring of every cross. His careful counting led him to be called the **Father of Genetics**.

TRAIT	DOMINANT FORM	RECESSIVE FORM
Plant height	Tall	Dwarf
Seed shape	Round	Wrinkled
Seed colour	Yellow	Green
Pod shape	Inflated	Constricted
Pod colour	Green	Yellow
Flower colour	Violet (purple)	White
Flower position	Axial	Terminal

The Monohybrid Cross

A **monohybrid cross** follows **one** pair of contrasting traits. Mendel crossed a pure tall plant (TT) with a pure dwarf plant (tt).

1. Parents (P)
TT (tall) × tt (dwarf). Gametes: T and t.

2. First generation (F_1)
All Tt, all tall. The dwarf trait has vanished, hidden by the dominant allele.

3. Self-pollinate F_1
Tt × Tt. Gametes: T and t from each parent.

4. Second generation (F_2)
Tall and dwarf plants appear in the ratio 3 : 1.

The Punnett square for Tt × Tt:

	T	T
T	TT	Tt
t	Tt	tt

RATIO IN F_2	RESULT
Phenotypic ratio	3 tall : 1 dwarf
Genotypic ratio	1 TT : 2 Tt : 1 tt



SIDEKICK TIP
Always write: parents → gametes → Punnett square → ratios. Examiners give marks for each step, so never jump to the answer.

Test cross

To find whether a tall plant is TT or Tt, cross it with a dwarf plant (tt). If any dwarf offspring appear, the tall parent was Tt. If all are tall, it was TT.

Mendel's Three Laws of Inheritance

1. 1. Law of Dominance
When two contrasting alleles are present together in a hybrid, only one (the dominant) is expressed. The other (the recessive) stays hidden.

2. 2. Law of Segregation (purity of gametes)
The two alleles of a gene separate during gamete formation, so each gamete carries only one allele. The alleles come together again, in random combinations, at fertilisation.

3. 3. Law of Independent Assortment
Alleles of different genes are distributed into gametes independently of each other. The inheritance of one trait does not affect the inheritance of another. This is seen in a dihybrid cross.

The Dihybrid Cross

A **dihybrid cross** follows **two** pairs of traits at once. Mendel crossed pea plants with round yellow seeds (RRYY) with plants having wrinkled green seeds (rryy). All F_1 offspring were RrYy: round and yellow. When F_1 plants were self-pollinated, each produced four types of gametes (RY, Ry, rY, ry) and 16 combinations were possible in F_2.

	RY	RY	RY	RY
RY	RRYY	RRYy	RrYY	RrYy
Ry	RRYy	RRyy	RrYy	Rryy
rY	RrYY	RrYy	rrYY	rrYy
ry	RrYy	Rryy	rrYy	rryy

F_2 PHENOTYPE	NUMBER OUT OF 16
Round yellow	9
Round green	3
Wrinkled yellow	3
Wrinkled green	1

KEY POINT

Phenotypic ratio of a dihybrid cross: 9 : 3 : 3 : 1. Two of the four classes (round green and wrinkled yellow) are new combinations that neither parent had. That proves the two genes are inherited independently. The syllabus asks only for the phenotypes in a dihybrid cross.

Sex Determination in Humans

Human males have the sex chromosomes **XY** and females **XX**. A woman produces only one kind of egg (22 + X). A man produces two kinds of sperm in equal numbers: half carry X and half carry Y.

	SPERM X	SPERM Y
Egg X	XX (girl)	XY (boy)
Egg X	XX (girl)	XY (boy)

- If an X-bearing sperm fertilises the egg, the baby is a **girl**. If a Y-bearing sperm does, the baby is a **boy**.
- The chance of each is 50 : 50 at every pregnancy.
- So **the father's sperm decides the sex of the child**, not the mother.

REAL LIFE

In some families, a mother is blamed for 'not having a son'. Genetics shows that the sex of the child depends on which sperm reaches the egg first, which is pure chance.

Sex-linked Inheritance

Genes located on a sex chromosome are **sex-linked**. The X chromosome is large and carries many genes that have nothing to do with sex, while the small Y carries very few. A male has only one X, so a recessive allele on it is always expressed. The syllabus covers only **X-linked recessive** diseases.

DISEASE	WHAT HAPPENS	CAUSE
Haemophilia	Blood does not clot properly; even a small cut bleeds for long	Recessive allele on the X chromosome for a clotting factor
Colour blindness (red-green)	Cannot tell red from green	Recessive allele on the X chromosome for the colour-sensing cone cells

Use X^H for the normal allele and X^h for the haemophilia allele (similarly for colour blindness, X^C and X^c).

GENOTYPE	FEMALE	MALE
X^HX^H	Normal	-
X^HX^h	Normal, but a **carrier**	-
X^hX^h	Affected (rare; lethal in the past)	-
X^HY	-	Normal
X^hY	-	Affected

Cross 1: carrier mother × normal father

	X^H (FATHER)	Y (FATHER)
X^H (mother)	X^HX^H normal girl	X^HY normal boy
X^h (mother)	X^HX^h carrier girl	X^hY affected boy

Out of four children: 1 normal girl, 1 carrier girl, 1 normal boy, 1 affected boy. **Half of the sons** are affected and half of the daughters are carriers. No daughter is affected.

Cross 2: affected father × normal mother

$X^HX^H \times X^hY$ gives X^HX^h (carrier daughters) and X^HY (normal sons). All daughters are carriers and all sons are normal. The affected father passes the allele to daughters only, never to sons, because sons get his Y.



KEY POINT

Sex-linked recessive diseases are more common in males because a male needs only one recessive allele to be affected, whereas a female needs two. The allele passes from an affected man, through his carrier daughter, to a grandson.



FUN FACT

Haemophilia was called the 'royal disease'. Queen Victoria was a carrier and passed it to several of her descendants in the royal houses of Europe.



BOARD ALERT

A carrier is **not** a patient. The female X^HX^h has normal blood clotting, but she can pass on the allele to half her children.

Variation and Mutation

- **Variation** means differences among individuals. It comes from crossing over, the random separation of chromosomes in meiosis, random fertilisation and mutation.
- **Mutation** is a sudden change in the DNA or in the number of chromosomes. Causes (mutagens) include X-rays, UV light, radioactive substances and some chemicals.

- Most mutations are harmful or neutral, but a few give an advantage, and these are the raw material for evolution.
- Only mutations in gametes or the cells that make them are passed on to the next generation.

HOT QUESTION

A colour-blind man marries a woman with normal vision whose father was colour blind. What are the chances that (a) their son and (b) their daughter will be colour blind? Show the cross.

HOT QUESTION

In pea plants, round seed (R) is dominant to wrinkled (r), and yellow (Y) is dominant to green (y). A plant with round yellow seeds is crossed with a plant with wrinkled green seeds and half the offspring are round yellow and half are wrinkled yellow. What is the genotype of the round yellow parent?

Mind Map Recap

BRANCH	WHAT TO REMEMBER
Terms	Gene, allele, dominant, recessive, homozygous, heterozygous, genotype, phenotype
Monohybrid	TT × tt → F1 Tt → F2 3:1 phenotype, 1:2:1 genotype
Dihybrid	RRYY × rryy → F2 9:3:3:1 phenotype
Mendel's laws	Dominance, segregation, independent assortment
Sex	XX female, XY male; sperm decides sex; 50:50
X-linked	Haemophilia, colour blindness; carrier mothers, affected sons

Practice Questions

#	QUESTION	ANSWER
1	Define phenotype and genotype with one example.	Phenotype is the observable trait (tall); genotype is the gene make-up (Tt).
2	A heterozygous tall plant is self-pollinated. Give the genotypic and phenotypic ratios of the offspring.	Genotypic 1 TT : 2 Tt : 1 tt; phenotypic 3 tall : 1 dwarf.
3	State Mendel's law of segregation.	The two alleles of a pair separate from each other during gamete formation so each gamete receives only one allele.
4	What is the phenotypic ratio in the F2 generation of a dihybrid cross?	9 : 3 : 3 : 1.
5	Which parent determines the sex of a child? Explain.	The father, because he produces X and Y sperm whereas the mother produces only X eggs.
6	Why is haemophilia more common in males?	The gene is on the X chromosome. A male has one X, so one recessive allele causes the disease; a female needs two.
7	Can a haemophilic father pass the disease to his son? Explain.	No. The son receives the father's Y chromosome, not his X, so the allele is passed only to daughters (as carriers).
8	What is a test cross and what is its purpose?	Crossing an individual showing the dominant trait with a homozygous recessive to find out whether it is homozygous or heterozygous.
9	Differentiate between homozygous and heterozygous.	Homozygous: two identical alleles (TT or tt). Heterozygous: two different alleles (Tt).

04

CHAPTER · BIOLOGY

Absorption by Roots

A tall tree lifts hundreds of litres of water from the soil every day, with no pump. This chapter explains how water gets into the root, why it moves, and what lifts it up to the leaves.



IN THIS CHAPTER

- Diffusion
- Osmosis and imbibition
- Turgor and plasmolysis
- Root hair
- Active and passive transport
- Ascent of sap

Three Ways Substances Move

PROCESS	DEFINITION	EXAMPLE
Diffusion	Movement of molecules from a region of higher concentration to a region of lower concentration, until evenly spread. No membrane needed.	Smell of perfume spreading; oxygen entering a leaf
Osmosis	Movement of water from a region of higher water concentration (dilute solution) to lower water concentration (concentrated solution) through a **semi-permeable (selectively permeable) membrane**.	Water entering a root hair
Imbibition	Absorption of water by solid, water-attracting substances (colloids such as cellulose, starch, proteins) without dissolving them, which makes them swell.	Dry seeds swelling; wooden door jamming in the monsoon; raisins in water



SIDEKICK TIP

Osmosis is a special case of diffusion in which only water moves, and a membrane is compulsory. Imbibition needs no membrane, only a solid that attracts water.

Osmosis in a cell

A plant cell behaves like a tiny osmotic system. The cell membrane is the semi-permeable membrane, and the cell sap in the vacuole is the solution. The strength with which a solution draws water in is described by its **osmotic pressure**.



KEY POINT

Osmotic pressure is the pressure that must be applied to a solution to just stop water from entering it through a semi-permeable membrane. The greater the concentration of the solution, the greater its osmotic pressure.

TERM	MEANING
Endosmosis	Entry of water into a cell when the outside solution is more dilute (hypotonic). The cell swells.
Exosmosis	Loss of water from a cell when the outside solution is more concentrated (hypertonic). The cell shrinks.
Turgidity	The firm, swollen state of a cell fully filled with water. The cell sap pushes the membrane against the wall. The wall pushes back with **wall pressure** (turgor pressure).
Flaccidity	The limp state of a cell that has lost water. Leaves and stems droop, a plant wilts.
Plasmolysis	Shrinking of the protoplasm away from the cell wall when a cell is placed in a strong (hypertonic) solution, because of exosmosis.
Deplasmolysis	Recovery of a plasmolysed cell when it is put back in pure water or a dilute solution. Water re-enters and the protoplasm swells to fill the cell.

REAL LIFE

Salting a pickle or sprinkling sugar on fruit draws water out of the cells by exosmosis, and out of any microbes too. That is why salt and sugar preserve food. Over-fertilised soil can plasmolyse root cells and kill a plant.

Experiments on Diffusion and Osmosis

1. Diffusion (KMnO4 in water)

Drop a crystal of potassium permanganate into a beaker of still water. Purple colour spreads outward from the crystal and eventually the whole water turns uniformly pink-purple. This shows diffusion of solute particles from high to low concentration.

2. Thistle funnel experiment

Tie a piece of egg membrane or parchment over the mouth of a thistle funnel filled with sugar solution and invert it in a beaker of water. The level of the solution rises in the stem as water enters through the membrane. This shows osmosis.

3. Potato osmoscope

Scoop a cavity in a peeled potato, put sugar solution in it and stand the potato in a dish of water. Mark the level. After some hours the level rises because water moves into the cavity. A boiled potato (dead membrane) shows no rise.

4. Potato in strong solution

A hollow in a potato filled with water and placed in strong sugar solution shows the level fall (exosmosis).

BOARD ALERT

A boiled potato is the control for the osmoscope. Boiling kills the cells, destroys the selective membrane, and osmosis stops. Without a control the conclusion is not valid.

The Root Hair: Built to Absorb

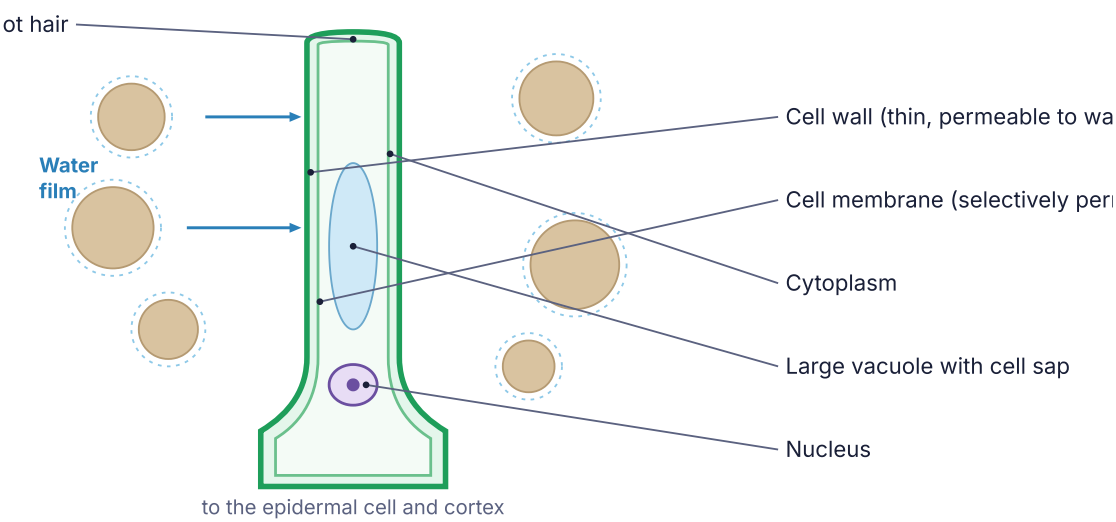


FIG 4.1 A SINGLE FULL-GROWN ROOT HAIR (SIMPLIFIED)

- It is a **tubular outgrowth of a single epidermal cell** near the root tip, in the root hair zone.
- The thin wall and membrane give a very short path for water.

- Thousands of root hairs give a **huge surface area** and press close to soil particles.
- The cell sap is more concentrated than the soil water, so water enters by osmosis.
- Root hairs live only a few days. New ones form as the root grows.

Features that make roots suitable for absorption

- Root hairs greatly increase the absorbing surface.
- The root grows towards moist soil (positive hydrotropism).
- Roots branch and spread widely through the soil.
- The root cap protects the delicate tip and the growing point as it pushes through the soil.
- The vascular tissue (xylem) in the centre carries the water upward.

From Soil to Xylem

1. Soil to root hair
Water in the soil is more dilute than the root hair's cell sap. It enters the root hair by osmosis. Dissolved mineral ions are absorbed by diffusion and active transport.

2. Across the cortex
The root hair cell becomes more dilute, so water passes to the next cell, which has stronger sap, and so on through the cortex cells, down a gradient of cell sap concentration. It also travels through cell walls.

3. Endodermis and pericycle
Water crosses the endodermis, a layer with a waterproof band, and the pericycle.

4. Into the xylem
Water finally enters the xylem vessels at the centre of the root and is lifted up the stem.

Absorption of Minerals: Active and Passive Transport

	PASSIVE TRANSPORT	ACTIVE TRANSPORT
Direction	Along the concentration gradient (high to low)	Against the concentration gradient (low to high)
Energy	Not needed	Needed, as ATP from respiration
Carrier proteins	Not needed (diffusion)	Needed
Example	Diffusion of ions when soil is rich in minerals	Uptake of nitrate and potassium ions when soil is poor in them



FUN FACT

Waterlogged soil has little oxygen. Roots cannot respire well, make less ATP and so cannot actively absorb minerals. That is why overwatered plants can wilt in spite of standing in water.

Forces Responsible for the Ascent of Sap

Water must go up a stem that can be over 100 metres tall. No single force can do this. Three forces act together.

FORCE	HOW IT WORKS
Root pressure	Active absorption of minerals makes the xylem sap concentrated, so water enters the xylem by osmosis and builds a push from below. It can raise water only a few metres. It is strongest at night and in the wet season. It causes guttation and bleeding.
Capillarity	Water rises in the very narrow xylem tubes because of adhesion between water and the vessel walls. It can lift water only a short distance.
Transpiration pull	Water evaporating from leaves creates suction on the water column in the xylem, like sucking a straw. The main force in tall trees.

Two properties of water make the pull work. **Cohesion** is the attraction between water molecules, which keeps the water column unbroken. **Adhesion** is the attraction between water and the xylem walls, which helps the column cling and not slip down. A continuous, unbroken column of water stretches from root to leaf.



KEY POINT

Cohesion + adhesion + transpiration pull is the best-accepted explanation for the ascent of sap, called the cohesion-tension (or transpiration pull) theory.

Experiment: water travels through the xylem

Place a leafy balsam twig with a freshly cut end in water containing eosin or red ink, and leave it in sunlight for an hour or two. Cut thin transverse and longitudinal sections of the stem. Only the xylem is stained red. The stain shows that water moves upward through the **xylem**, not the phloem or the cortex.



HOT QUESTION

A farmer applies a very large amount of chemical fertiliser to a field and the crop plants wilt next day, even though the soil is wet. Give a reason using the terms osmosis, concentration and exosmosis.

Mind Map Recap

BRANCH	WHAT TO REMEMBER
Movement	Diffusion (no membrane), osmosis (water, semi-permeable membrane), imbibition (solid colloids swell)
Cell states	Turgid, flaccid, plasmolysed; deplasmolysis reverses
Root hair	Single epidermal cell; large surface; thin wall; more concentrated cell sap
Mineral uptake	Passive (along gradient) and active (against gradient, ATP)
Ascent of sap	Root pressure + capillarity + transpiration pull; cohesion and adhesion

Practice Questions

#	QUESTION	ANSWER
1	Define osmosis.	Movement of water from a region of higher water concentration to lower through a semi-permeable membrane.
2	What is the difference between diffusion and imbibition?	Diffusion is the spreading of molecules from high to low concentration. Imbibition is absorption of water by solid colloids that swell.
3	What happens to a plant cell placed in a concentrated salt solution? Name the process.	It loses water by exosmosis, the protoplasm shrinks away from the wall; this is plasmolysis.
4	State two ways in which a root hair is adapted for absorption.	It has a large surface area and a thin wall; its cell sap is more concentrated than soil water.
5	Differentiate between active and passive transport.	Passive uses no energy and goes down a gradient; active uses ATP and goes against a gradient.
6	Why does a plant wilt on a hot day even when soil has water?	Transpiration exceeds absorption, cells lose turgor and become flaccid.
7	Name the three forces that lift water in a tall tree. Which is the most important?	Root pressure, capillarity, transpiration pull. Transpiration pull is the most important.
8	Why is a boiled potato used as a control in the potato osmoscope?	Boiling kills the cells so membranes lose selective permeability; no osmosis occurs, proving that the rise in the living potato is due to osmosis.

05

CHAPTER · BIOLOGY

Transpiration

Plants drink a lot and keep very little. More than nine-tenths of the water a plant absorbs leaves through its leaves as vapour. This chapter explains why, how it is controlled, how we measure it and how plants limit the loss.



IN THIS CHAPTER

- Meaning and types
- Stomatal mechanism
- Factors
- Ganong's potometer
- Experiments
- Guttation and bleeding

What Is Transpiration?

KEY POINT

Transpiration is the loss of water in the form of vapour from the aerial parts of a plant, mainly through the stomata of the leaves.

Of all the water absorbed by the roots, only a tiny fraction (around one or two per cent) is used for photosynthesis and growth. The rest, more than 90 per cent, is lost by transpiration. It is called a **necessary evil**: it wastes water, but it also does useful work.

TYPE	THROUGH	SHARE OF TOTAL	NOTES
Stomatal	Stomata on the leaf (mostly the lower surface)	About 80 to 90%	The main type; day only; controlled by guard cells
Cuticular	The waxy cuticle covering leaf and stem	About 10%	Greater when the cuticle is thin
Lenticular	Lenticels, small pores in the bark of woody stems	Less than 1%	Slight amount, continues at night

Importance of transpiration

- **Ascent of sap:** the pull it creates lifts water and minerals from the root to the leaves.
- **Cooling:** evaporation takes up heat, so leaves are kept from overheating in the sun.
- **Supply of water:** a constant stream of water brings dissolved minerals to the leaf cells.
- **Turgidity:** keeps cells firm, and maintains the water balance within the plant.
- **Humidity and rain:** forests add large amounts of water vapour to the air.

BOARD ALERT

Do not call transpiration the same as evaporation. Evaporation is a purely physical process from any wet surface. Transpiration is a **physiological process** in a living plant; it is controlled by guard cells and by the plant's water status.

Opening and Closing of Stomata: The Potassium Ion Exchange Theory

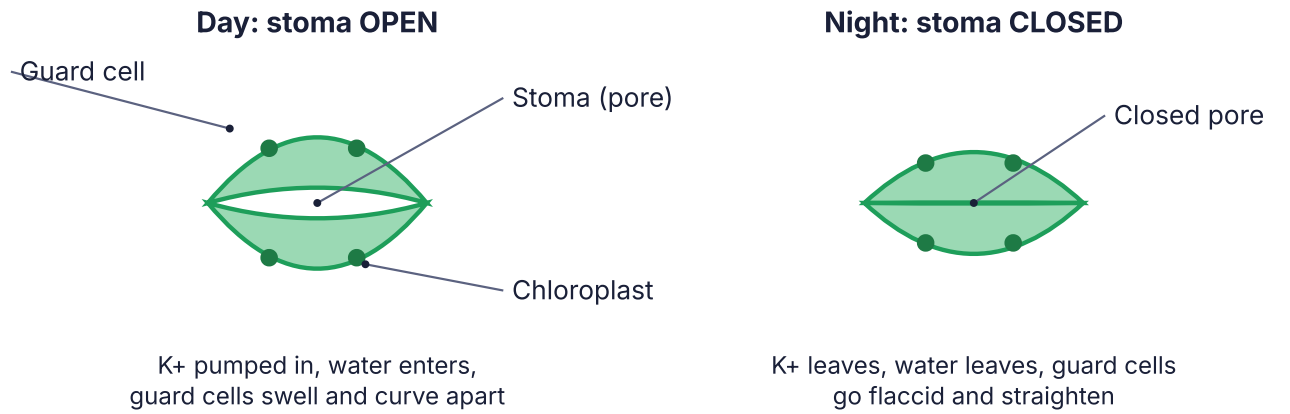


FIG 5.2 STOMATA: OPEN AND CLOSED (SURFACE VIEW)

Each stoma is a pore between two bean-shaped **guard cells**. The inner wall of each guard cell is thick and the outer wall is thin, so when the cells swell with water, they bend outward and the pore opens.

1. Morning: opening

In light, guard cells photosynthesise and make ATP. The ATP drives the pumping of potassium ions (K^+) into the guard cells from the neighbouring epidermal cells.

2. Water follows

The guard cell sap becomes more concentrated. Water enters from the neighbouring cells by osmosis. The guard cells turn turgid and bow outward. The stoma opens.

3. Evening: closing

In the dark or when the plant is short of water, K^+ moves back out into the neighbouring cells. Water follows by osmosis, the guard cells become flaccid and the stoma closes.

SIDEKICK TIP

K^+ in, water in, stoma opens. K^+ out, water out, stoma closes. It is the movement of potassium, then of water, that opens and closes the pore.

Factors Affecting the Rate of Transpiration

FACTOR	EFFECT ON RATE	REASON
Light	Increases	Stomata open in light; leaf warms
Temperature	Increases with a rise	Faster evaporation; air holds more vapour
Humidity	Decreases when humidity is high	Smaller difference in vapour between leaf and air
Wind	Increases	Removes vapour near the leaf, keeping the gradient steep (very strong wind may close stomata)
Soil water	Decreases when soil is dry	Less water available; stomata close
Atmospheric pressure	Increases when low	Water vapour diffuses faster at low pressure
Number and position of stomata, leaf area, cuticle, leaf hairs	More stomata and larger area: more. Thick cuticle, hairs: less	Internal factors that depend on the plant

Measuring Transpiration: Ganong's Potometer

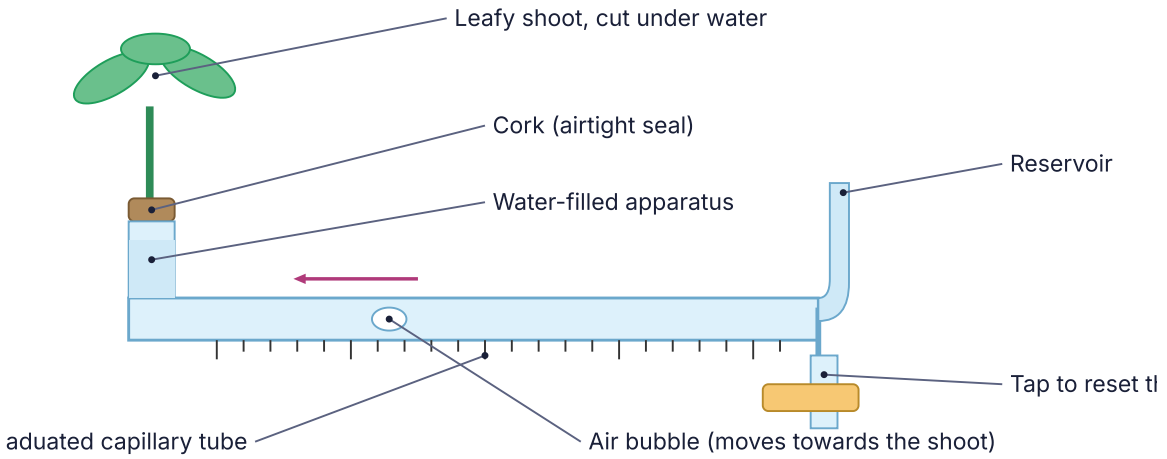


FIG 5.1 GANONG'S POTOMETER (SIMPLIFIED)

The potometer actually measures the **rate of water uptake** by a cut shoot, which is used as a close estimate of the rate of transpiration. The distance moved by the air bubble in a set time shows the water absorbed. The rate can be compared under conditions such as sun, shade, wind and fan.

Limitations of the potometer

- It measures water **absorbed**, not water **lost**. A little water is used in photosynthesis and some is stored.
- A cut shoot has no roots, so it does not behave exactly like an intact plant.
- Any air bubble or leak in the joints gives false readings.

- Conditions in the lab differ from those in the field.
- Only a single shoot is tested at a time.

Experiments on Transpiration

1. Loss in weight (leafy shoot in a test tube)

Put a leafy shoot in a test tube of water. Pour a layer of oil on top of the water to stop direct evaporation, and plug the mouth with cotton wool. Weigh the whole set-up and weigh it again after a few hours. The loss of weight is the water lost by transpiration. A similar test tube of water without a shoot acts as the control.

2. Cobalt chloride paper (unequal transpiration in a dorsiventral leaf)

Dry cobalt chloride paper is **blue** and turns **pink** when it absorbs moisture. Fix a strip on the upper surface and another on the lower surface of a leaf of a potted plant, cover each with a glass slide and clip them together. Note the time for each to turn pink. The paper on the **lower** surface turns pink first, because a dorsiventral leaf has many more stomata on its lower surface.



KEY POINT

Cobalt chloride paper experiment proves that **transpiration is greater from the lower surface** of a dorsiventral leaf, since it has more stomata.

The bell jar experiment is described in the Practical Lab Companion at the end of this book.

Adaptations to Reduce Transpiration

ADAPTATION	EXAMPLE
Thick waxy cuticle	Cactus, Nerium (oleander)
Sunken stomata, stomata only on the lower surface	Nerium, pine
Leaves reduced to spines or small scales	Opuntia, *Casuarina*
Rolling of leaves in dry weather	Grasses such as *Ammophila*
Hairs on the leaf surface that trap moist air	Many desert plants
Fleshy stems storing water; photosynthesis in the stem	Opuntia
Shedding leaves in the dry season	Teak, peepal

Guttation and Bleeding

	GUTTATION	BLEEDING
What it is	Loss of water droplets from the tips and edges of leaves	Oozing of sap from a cut or injured stem
Through	Hydathodes (water pores at the vein ends)	The wound
Caused by	Root pressure	Root pressure
When	Night or early morning when humidity is high and transpiration is low	Whenever the plant has strong root pressure
Liquid	Water with some dissolved salts and sugars	Sap containing water, minerals and organic substances
Examples	Grass, tomato, colocasia, strawberry	Cut stem of grapevine or tomato



FUN FACT

The drops you see on blades of grass in the morning are not all dew. Some of them are guttation drops pushed out by root pressure.



REAL LIFE

In summer, the temperature under a tree is clearly lower than in the open. A large part of this is transpiration cooling the leaves and the air around them.



HOT QUESTION

A potted plant is placed under a glass bell jar on a sunny window sill, and drops of water appear on the inside of the jar. Why must the soil surface be covered with polythene, and what control is needed?



HOT QUESTION

The stomata of desert plants like cacti stay closed during the day. How can this plant still take in carbon dioxide, and what advantage does it gain?

Mind Map Recap

BRANCH	WHAT TO REMEMBER
Definition	Loss of water vapour from aerial parts; stomatal 80-90%, cuticular ~10%, lenticular tiny
Stomatal control	K+ moves in, water follows, guard cells turgid, stoma opens
Factors	Light, temperature, humidity, wind, soil water
Measure	Ganong's potometer (water uptake); cobalt chloride paper; weight loss
Adaptations	Thick cuticle, sunken stomata, spines, hairs, rolling
Guttation / bleeding	Both caused by root pressure; hydathodes vs wound

Practice Questions

#	QUESTION	ANSWER
1	Define transpiration and name its three types.	Loss of water vapour from aerial parts of a plant. Stomatal, cuticular, lenticular.
2	State two benefits of transpiration.	It pulls water and minerals up the plant and cools the leaf.
3	Explain how stomata open in the morning.	Light stimulates guard cells to make ATP; K+ enters from neighbouring cells, sap concentration rises, water enters by osmosis, guard cells become turgid and the pore opens.
4	What is the function of the oil layer in the weight-loss experiment?	It prevents evaporation of water directly from the test tube, so any loss of weight is due to transpiration only.
5	Why does cobalt chloride paper on the lower surface turn pink first?	There are more stomata on the lower surface so more water vapour is lost there.
6	State two limitations of Ganong's potometer.	It measures water absorption, not loss; a cut shoot without roots does not represent a whole plant; air leaks affect the readings.
7	Differentiate between guttation and transpiration.	Guttation loses liquid water through hydathodes at night because of root pressure; transpiration loses water vapour mainly through stomata by day.
8	Name three adaptations of desert plants to reduce transpiration.	Thick cuticle, leaves reduced to spines, sunken stomata, fleshy water-storing stems.
9	Why does transpiration increase on a windy day?	Wind sweeps away the water vapour near the leaf surface, maintaining a steep vapour gradient.

06

CHAPTER · BIOLOGY

Photosynthesis

Every breath you take and every meal you eat depends on one reaction that happens in green leaves. Photosynthesis turns light, water and air into food and oxygen. Learn it as a short story with two acts.



IN THIS CHAPTER

- Equation
- Chloroplast
- Light reaction
- Dark reaction
- Experiments
- Carbon cycle

What Is Photosynthesis?

KEY POINT

Photosynthesis is the process by which green plants use light energy, absorbed by chlorophyll, to build carbohydrates (glucose) from carbon dioxide and water, releasing oxygen.

$$6CO_2 + 12H_2O \rightarrow C_6H_{12}O_6 + 6O_2 + 6H_2O$$

(light energy, chlorophyll)

The water on the product side is new water made in the process. Cancelling the same number of water molecules on both sides gives the form used in most books: **$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$**. Both are correct.

RAW MATERIAL	SOURCE
Carbon dioxide	From the air through the stomata
Water	From the soil through the roots and xylem
Light energy	Sun, absorbed by chlorophyll
Chlorophyll	The green pigment in chloroplasts

The glucose is changed to starch for storage, to cellulose for cell walls, and to proteins and fats with the help of minerals. The oxygen is released through the stomata.

Where It Happens: The Chloroplast

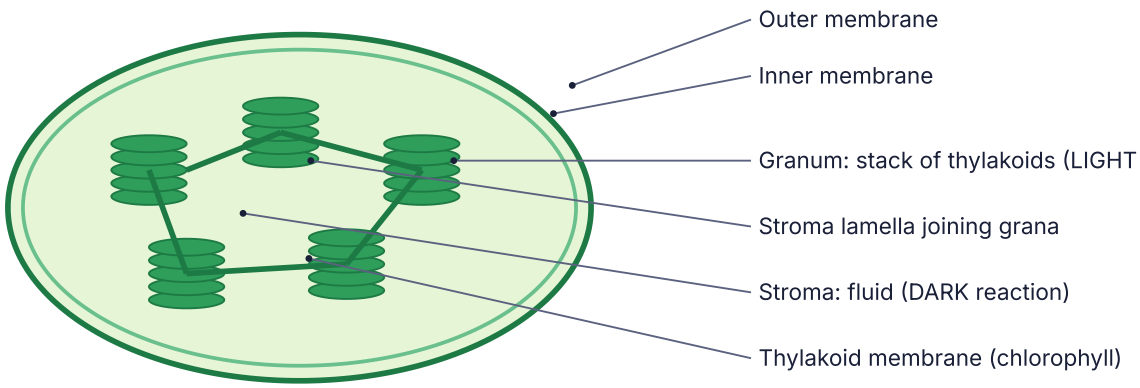


FIG 6.1 INTERNAL STRUCTURE OF A CHLOROPLAST (SIMPLIFIED)

A chloroplast is a double-membraned organelle. Inside is a fluid **stroma** that contains stacks of coin-like **thylakoids** called **grana**, joined by stroma lamellae. Chlorophyll is embedded in the thylakoid membranes. The leaf also contains yellow-orange pigments (carotenoids), but chlorophyll absorbs mainly red and blue light and reflects green, which is why leaves look green.

The Two Phases of Photosynthesis

	LIGHT REACTION (PHOTOCHEMICAL PHASE)	DARK REACTION (BIOSYNTHETIC PHASE)
Site	Grana (thylakoid membranes)	Stroma
Needs light?	Yes, directly	Not directly, but uses products of the light reaction
Main events	Chlorophyll activated; photolysis of water; ATP and NADPH formed; oxygen released	Carbon dioxide combines with hydrogen from NADPH to form glucose; uses ATP
Products	ATP, NADPH, O_2	Glucose (sugar), $NADP^+$, ADP

Light reaction, step by step

1. Activation of chlorophyll
Chlorophyll absorbs light. Its electrons gain energy and are excited.

2. Photolysis of water
Light splits water: $2H_2O \rightarrow 4H^+ + 4e^- + O_2$. The oxygen released comes from the water, not from CO_2.

3. Formation of ATP
Part of the energy of the excited electrons is used to join ADP and phosphate to make ATP. This is called photophosphorylation.

4. Formation of NADPH
The hydrogen released from water is picked up by the carrier $NADP^+$ to make NADPH.

Dark reaction, in brief

In the stroma, the hydrogen carried by NADPH, together with energy from ATP, is added to carbon dioxide and converted to **glucose**. This stage can go on in the light as well. Detailed equations are not needed.



SIDEKICK TIP

Light reaction makes the tools (ATP, NADPH) and oxygen. Dark reaction uses the tools to make sugar.
Remember: grana = light, stroma = dark.

Adaptations of a Leaf for Photosynthesis

- Broad, thin, flat lamina: a large area for light and a short path for gases.
- Large numbers of chloroplasts in the palisade cells just below the upper surface, where light is strongest.
- Stomata for gas exchange, usually more on the lower surface.
- Network of veins: xylem brings water, phloem takes sugar away.
- Loose spongy mesophyll with air spaces for quick diffusion of CO_2.
- Leaves arranged in a leaf mosaic so that they do not shade each other. Thin transparent cuticle and epidermis.

Factors that Affect the Rate

FACTOR	EFFECT
Light intensity	Rate rises with light until another factor becomes limiting
CO_2 concentration	Rate rises with CO_2 (air has only about 0.04%)
Temperature	Best around 25 to 35 degrees Celsius; falls when too cold or too hot (enzymes slow or denature)
Water	Severe shortage closes stomata and reduces the rate
Chlorophyll	No chlorophyll, no photosynthesis

Experiments on Photosynthesis

Experiments begin with a **destarched** plant, one whose stored starch has been used up, so that any starch found later was made during the experiment.

1. Destarch
Keep the potted plant in complete darkness for 24 to 48 hours. The plant uses up its starch for respiration.

2. Set up
Expose the plant to the condition being tested (light, CO2, chlorophyll) alongside a control.

3. Kill the leaf
Dip the leaf in boiling water for about a minute. This stops all chemical reactions and softens the leaf.

4. Remove chlorophyll
Boil the leaf in methylated spirit in a water bath. The leaf turns pale and the green colour passes into the spirit.

5. Wash
Rinse the leaf in warm water to soften it.

6. Test
Spread the leaf in a petri dish and add a few drops of iodine solution. A blue-black colour means starch is present.



BOARD ALERT

Never heat methylated spirit directly on a flame. Always use a water bath, because the spirit is highly inflammable.

EXPERIMENT	SET-UP	RESULT	CONCLUSION
Necessity of light	Part of a destarched leaf covered with black paper strips (or a stencil)	Covered part stays brown, uncovered part turns blue-black	Light is necessary
Necessity of CO_2	Destarched plant in a bell jar with a dish of potassium hydroxide (KOH, absorbs CO_2); control jar with water	Leaf with KOH stays brown, control turns blue-black	CO_2 is necessary
Necessity of chlorophyll	Variegated leaf (green and white patches), as in *Coleus* or croton	Only the green parts turn blue-black	Chlorophyll is necessary
Release of oxygen	*Hydrilla* or *Elodea* under an inverted funnel and test tube in sunlight	Gas bubbles collect in the test tube. A glowing splinter relights	Oxygen is released



SIDEKICK TIP

A pinch of sodium bicarbonate in the water of the *Hydrilla* experiment gives off extra CO_2 and speeds up the bubbling. In the dark, or with a boiled plant, no bubbles appear.

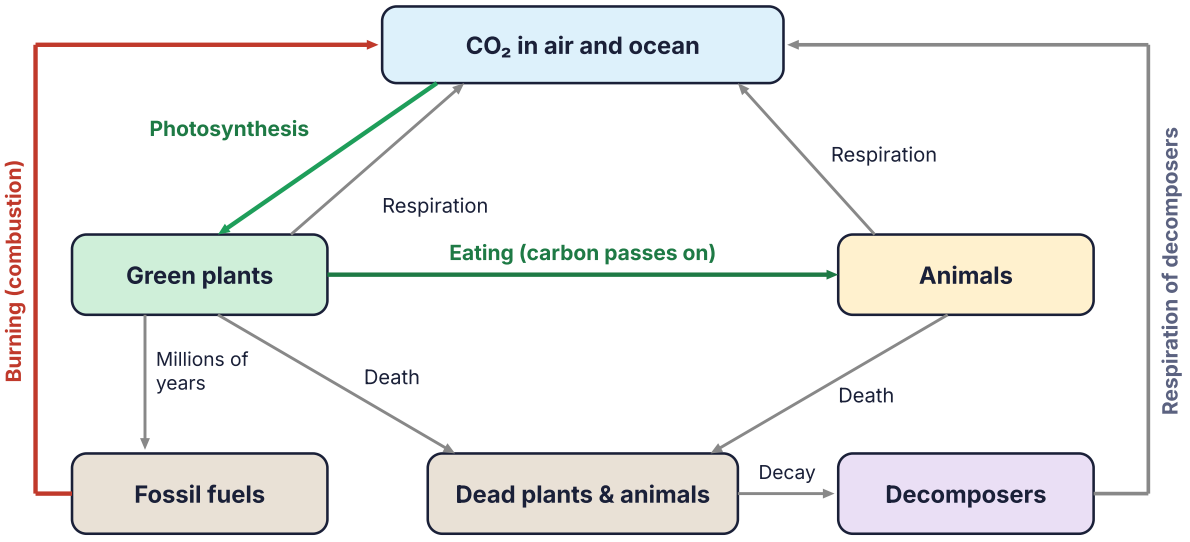
Opening and Closing of Stomata

Photosynthesis depends on open stomata to admit CO_2. As explained in the last chapter, stomata open when K^+ ions flow into the guard cells and water follows by osmosis, and they close when K^+ flows out.

Importance of Photosynthesis

- It is the **source of all food** and of the energy in food chains.
- It releases **oxygen** for the respiration of living things.
- It removes carbon dioxide from the air and helps balance the carbon cycle.
- Coal, petroleum and natural gas are the remains of ancient photosynthetic life.
- It converts light energy into chemical energy.

The Carbon Cycle



Green arrows take carbon in; grey and red arrows return carbon to the air.

FIG 6.2 THE CARBON CYCLE

Carbon moves between the air, living things and the ground in a continuous loop. Plants take CO_2 in by photosynthesis. Carbon passes to animals when they eat plants. Respiration by plants, animals and decomposers returns CO_2 to the air. Dead organisms are decomposed. Burning of wood and fossil fuels also adds CO_2.

REAL LIFE

The extra carbon dioxide we release by burning coal, oil and gas is being added faster than plants and oceans can remove it. That is the main reason for global warming.

HOT QUESTION

A potted plant is kept in a dark room for 48 hours. One of its leaves is partly covered by black paper and the whole plant is then placed in sunlight for 4 to 6 hours. Predict the iodine test result on that leaf, explain, and say why we destarched the plant.

HOT QUESTION

Aquatic plants release more oxygen bubbles in bright light. A student argues that the oxygen comes from the CO_2 absorbed. Which step of photosynthesis shows that this is wrong?

Mind Map Recap

BRANCH	WHAT TO REMEMBER
Equation	6CO2 + 6H2O → C6H12O6 + 6O2 (light, chlorophyll)
Chloroplast	Grana = light phase; stroma = dark phase
Light phase	Photochemical: activation, photolysis of water, ATP + NADPH, O2 out
Dark phase	Biosynthetic: CO2 + hydrogen from NADPH → glucose
Experiments	Destarch, boil, spirit, iodine test
Carbon cycle	Photosynthesis in; respiration, decay, burning out

Practice Questions

#	QUESTION	ANSWER
1	Write the balanced overall equation for photosynthesis.	6CO2 + 6H2O gives C6H12O6 + 6O2 in the presence of light and chlorophyll.
2	Where in the chloroplast do the light and the dark reactions occur?	Light reaction in the grana; dark reaction in the stroma.
3	What is photolysis of water?	Splitting of water by light energy into hydrogen ions, electrons and oxygen.
4	Name the products of the light reaction.	ATP, NADPH and oxygen.
5	Why is a plant destarched before an experiment?	To remove existing starch so that any starch found later was made during the experiment.
6	Why is methylated spirit heated in a water bath?	It is highly inflammable, so it must not be heated directly on a flame.
7	What is the role of KOH in the CO2 experiment?	It absorbs CO2 from the air in the jar, so the plant has none for photosynthesis.
8	How does the opening of stomata help photosynthesis?	It allows CO2 to diffuse into the leaf and oxygen to leave.
9	Name two ways in which carbon returns to the atmosphere.	Respiration and combustion (also decay).

07

CHAPTER · BIOLOGY

Chemical Coordination in Plants

Plants have no nerves and no brain, yet they bend towards light, send roots down and ripen fruit on time. The messengers are chemicals called plant hormones. This chapter introduces the five groups and the movements they control.



IN THIS CHAPTER

- Plant growth regulators
- Auxins
- Gibberellins
- Cytokinins
- Abscisic acid and ethylene
- Tropisms

Plant Growth Regulators

Plant hormones (**phytohormones** or plant growth regulators) are chemical substances made in one part of the plant and used, in very small amounts, to control growth and responses in another part. Some promote growth, and others inhibit it.

HORMONE	MADE IN	MAIN EFFECTS	TYPE
Auxins (e.g. IAA)	Shoot tip, young leaves	Cell elongation; phototropism and geotropism; apical dominance; root formation on cuttings; fruit growth without fertilisation (parthenocarpy); weed control (2,4-D)	Promoter
Gibberellins	Young leaves, embryo, roots	Stem elongation (tall plants, bolting); breaks seed and bud dormancy and aids germination; larger seedless grapes; flowering in some plants	Promoter
Cytokinins	Root tips, developing fruit and seeds	Promote cell division; delay ageing (senescence) of leaves; growth of side buds; tissue culture	Promoter
Abscisic acid (ABA)	Mature leaves, roots, fruit	Closes stomata in water shortage; keeps seeds and buds dormant; leaf and fruit fall; 'stress hormone'	Inhibitor
Ethylene	Ripening fruit, ageing tissues	Ripens fruit; leaf and fruit fall; promotes flowering in pineapple; a **gas**	Mainly inhibitor / ripening

SIDEKICK TIP

Auxin = elongating cells, Gibberellin = Growing tall, Cytokinin = Cell division, ABA = Stress and sleep, Ethylene = Ripening gas. One keyword per hormone is enough to answer most 2-mark questions.

Apical dominance

The shoot tip makes auxin, which moves down and stops the buds in the leaf axils from growing. If the tip is cut off, as when a gardener trims a hedge, the side buds grow and the plant becomes bushy. Cytokinins from the roots favour side buds, so the balance between these two hormones decides the shape of the plant.

REAL LIFE

Raw bananas ripen faster when kept in a bag with a ripe banana or an apple, because ripe fruit gives off ethylene gas. Fruit sellers use the same trick, and one rotten fruit can spoil the rest.

FUN FACT

Gibberellin was discovered from a fungus (*Gibberella*) that caused 'foolish seedling' disease in rice. The infected plants grew far taller than normal and then fell over.

Tropic Movements

A **tropism** is a directional growth movement of a plant part in response to a stimulus. If the part grows **towards** the stimulus it is **positive**; if **away**, it is **negative**. These movements are slow, permanent growth curves, unlike the fast, temporary movement of the touch-me-not plant (a nastic movement).

TROPISM	STIMULUS	RESPONSE	EXAMPLE
Phototropism	Light	Shoot: positive. Root: negative	Sunflower stem or houseplant bending to the window
Geotropism	Gravity	Root: positive (grows down). Shoot: negative (grows up)	A seedling placed sideways: root turns down, shoot turns up
Hydrotropism	Water	Root: positive	Roots growing toward a leaking pipe or moist soil
Thigmotropism	Touch or contact	Tendrils: positive (coil around support)	Tendrils of pea, cucumber, gourd and grapevine
Chemotropism	Chemicals	Pollen tube: positive	Pollen tube growing down the style towards the ovule, attracted by chemicals from the ovule

How does a shoot bend towards light?

1. Detect
The tip senses the light direction. Charles Darwin showed that a seedling bent to light only if the tip was exposed. A covered tip failed to bend.

2. Redistribute
Auxin moves away from the lit side and collects on the shaded side.

3. Elongate
More auxin on the shaded side makes those cells elongate faster.

4. Bend
The shaded side grows longer and the shoot curves towards the light.

BOARD ALERT
Roots respond differently to the same hormone. The auxin level that boosts growth in a stem **slows** growth in a root. That is why roots are negatively phototropic and positively geotropic.

HOT QUESTION
A gardener finds that the potted plants near a window all lean outwards. She turns the pots around, and after a few days the plants bend the other way. Explain, in terms of auxin, why the plants bend, and name the tropism.

HOT QUESTION
A farmer sprays a broad-leaved weed killer that contains a synthetic auxin on a field of wheat. The weeds die but the wheat survives. How can a growth-promoting hormone kill the weeds?

Mind Map Recap

BRANCH	WHAT TO REMEMBER
Promoters	Auxin (elongation, tropisms, apical dominance), Gibberellin (stem growth, germination), Cytokinin (division, delays ageing)
Inhibitors	ABA (stress, dormancy, stomatal closure), Ethylene (ripening gas)
Tropisms	Photo-, geo-, hydro-, thigmo-, chemo-
Rule	Toward = positive; away = negative

Practice Questions

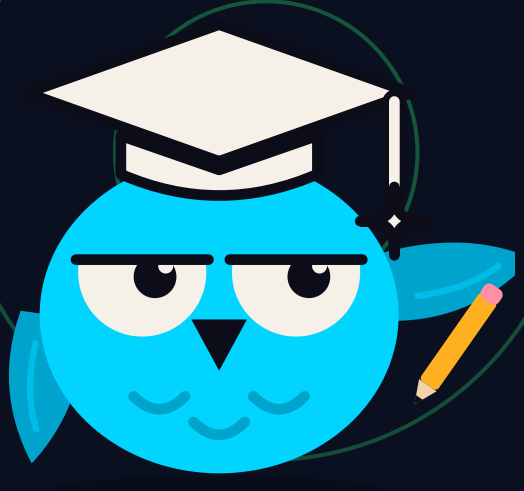
#	QUESTION	ANSWER
1	Name the plant hormone responsible for the bending of a shoot towards light.	Auxin.
2	State two functions of gibberellins.	Stem elongation and breaking of seed dormancy (also larger seedless fruit).
3	Which hormone is a gas? Give one use.	Ethylene; it ripens fruits.
4	Which hormone delays the ageing of leaves?	Cytokinin.
5	What is apical dominance?	Inhibition of growth of side buds by auxin produced in the shoot tip.
6	Name the tropism shown by (a) tendrils (b) a pollen tube.	(a) Thigmotropism (b) Chemotropism.
7	Why is ABA called the stress hormone?	It rises in drought, closes stomata and promotes dormancy, helping the plant survive stress.
8	Distinguish between positive and negative geotropism.	Roots growing towards gravity show positive geotropism; shoots growing away from it show negative geotropism.

08

CHAPTER · BIOLOGY

The Circulatory System

Every cell in your body needs a delivery service: oxygen and food in, wastes out. The heart, the vessels and the blood form that service. This is the longest chapter in the book. Take it in sections.



IN THIS CHAPTER

- Blood
- Clotting
- Heart
- Blood vessels
- Double circulation
- ABO and Rh
- Lymph
- Hepatic portal system

Blood: A Liquid Connective Tissue

An adult has about **5 litres** of blood. It is a fluid tissue made of a liquid, the **plasma**, and the cells floating in it. Plasma makes up about 55 per cent and the cells about 45 per cent.

COMPONENT	DETAILS	FUNCTION
Plasma	Pale yellow fluid; about 90% water; contains plasma proteins (albumin, globulins, fibrinogen), glucose, amino acids, salts, hormones, urea and dissolved gases	Carries digested food, hormones and wastes; helps in clotting; maintains osmotic balance
Red blood cells (RBC, erythrocytes)	Biconcave discs, no nucleus in mammals, contain haemoglobin; about 5 million per cubic millimetre (4.5 to 5.5 million); made in red bone marrow; live about 120 days	Carry oxygen as oxyhaemoglobin, and a little CO_2
White blood cells (WBC, leucocytes)	Colourless, have a nucleus, irregular in shape; 5,000 to 10,000 per cubic millimetre. Granulocytes (neutrophils, eosinophils, basophils) and agranulocytes (lymphocytes, monocytes)	Defence: engulf germs (phagocytosis) and make antibodies
Platelets (thrombocytes)	Tiny cell fragments without a nucleus; 1.5 to 4 lakh per cubic millimetre	Start blood clotting



FIG 8.1 MAIN TYPES OF BLOOD CELLS (NOT TO SCALE; WBCs ARE DRAWN LARGER THAN RBC FOR DETAIL)

Why mammalian RBCs have no nucleus (or other organelles)

- The space saved lets each cell hold **more haemoglobin**, so it can carry more oxygen.
- Without mitochondria, an RBC does not use up the oxygen it carries. It makes energy by anaerobic breakdown of glucose.
- The **biconcave shape** gives a larger surface for gas exchange and allows the cell to squeeze through narrow capillaries.

BOARD ALERT

Do not write that RBCs carry oxygen in the plasma. Oxygen is carried mostly in combination with haemoglobin as **oxyhaemoglobin**. Haemoglobin contains **iron**, so a lack of iron causes anaemia.

FUN FACT

Red blood cells are made at about two million per second in red bone marrow. Old ones are broken down in the spleen and liver, and the iron is recycled.

Blood Coagulation (Clotting)

When a vessel is cut, the blood must seal the wound quickly to stop bleeding and keep germs out.

1. Injury
Platelets and the damaged tissue release an enzyme, thromboplastin (thrombokinase).

2. Prothrombin to thrombin
In the presence of calcium ions, thromboplastin converts inactive prothrombin in the plasma into active thrombin. Prothrombin is made in the liver using vitamin K.

3. Fibrinogen to fibrin
Thrombin converts the soluble protein fibrinogen into insoluble threads of fibrin.

4. Clot
Fibrin threads form a net that traps blood cells. The clot dries to a scab and seals the wound.

KEY POINT

Prothrombin → thrombin → fibrinogen → fibrin. Calcium and vitamin K are essential. Without calcium or clotting factors, as in haemophilia, blood does not clot.

Tissue Fluid and Lymph

As blood passes through the capillaries, some plasma leaks out through their thin walls. This fluid, with water, dissolved food, oxygen and salts, but **without RBCs and large proteins**, bathes the cells. It is the **tissue fluid**, and it is the medium in which exchange between the blood and the cells takes place. Most of it goes back into the capillaries. The rest drains into tiny **lymph capillaries** and is then called **lymph**.

FEATURE	BLOOD	LYMPH
Colour	Red	Colourless or pale yellow
RBC	Present	Absent
WBC	Present	Mainly lymphocytes
Plasma proteins	More	Less
Movement	Pumped by the heart	Slow, by muscle movement and valves in lymph vessels; no pump
Flow	Round trip in a circuit	One way, from tissues to the veins near the heart

- **Functions of lymph:** returns tissue fluid to the blood; carries fats absorbed in the intestine (through lacteals in the villi); fights infection with lymphocytes; filtered in **lymph nodes** that also trap germs.

- The **lymphatic system** consists of lymph, lymph vessels, lymph nodes and lymphatic organs. Lymph finally empties into large veins in the neck region (the subclavian veins).

Lymphatic organs: spleen and tonsils

ORGAN	LOCATION	FUNCTIONS
Spleen	Upper left of the abdomen, behind the stomach. The largest lymphatic organ.	Destroys old and worn-out red blood cells; stores blood and releases it during emergencies such as heavy bleeding; makes lymphocytes and filters germs from the blood.
Tonsils	Masses of lymphoid tissue at the back of the throat, on either side of the pharynx.	Trap germs that enter by the nose and mouth and make lymphocytes: a first line of defence for the respiratory and digestive tracts.



SIDEKICK TIP

Spleen = Scrapyard. Old red blood cells are sent to the spleen to be broken down and recycled.



FUN FACT

A person can live without a spleen but is more prone to certain infections, because the body loses one of its blood filters.

The Heart

The heart is a hollow, muscular organ about the size of a fist, lying in the chest between the lungs and tilted slightly to the left. It is enclosed in a double membrane, the **pericardium**, with fluid between the layers that reduces friction. The heart muscle (cardiac muscle) is fed by the **coronary arteries**. A blockage in them causes a heart attack.

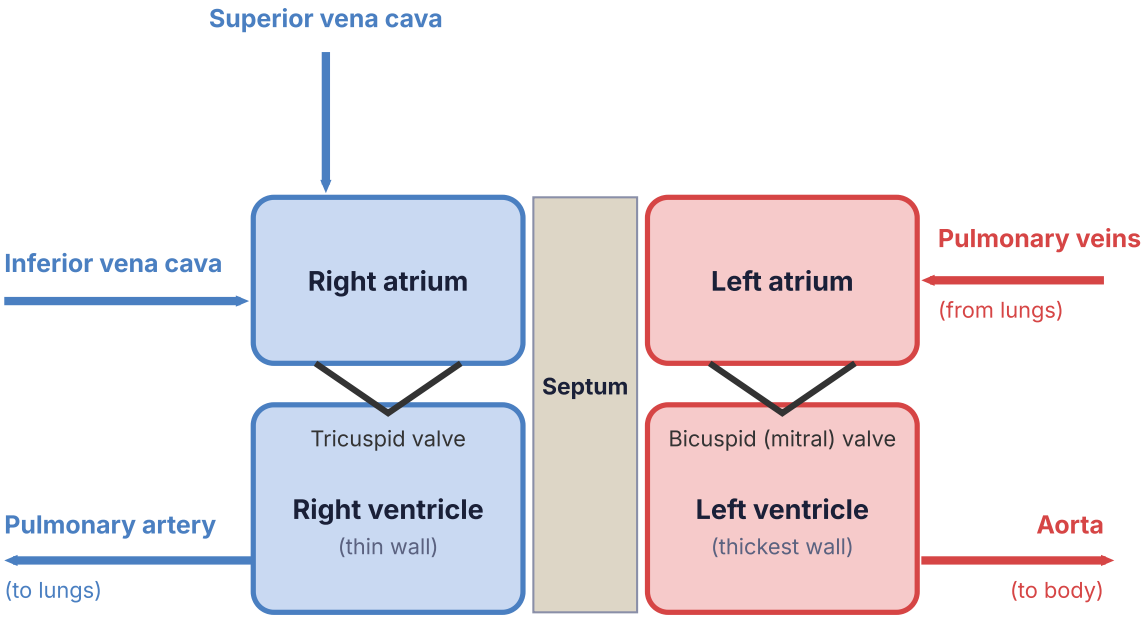


FIG 8.2 THE HEART IN SECTION (SCHEMATIC, VIEWED FROM THE FRONT)

CHAMBER	RECEIVES BLOOD FROM	SENDS BLOOD TO
Right atrium	Superior and inferior vena cava (deoxygenated blood from the body)	Right ventricle, through the tricuspid valve
Right ventricle	Right atrium	Lungs, through the pulmonary artery
Left atrium	Pulmonary veins (oxygenated blood from the lungs)	Left ventricle, through the bicuspid valve
Left ventricle	Left atrium	The whole body, through the aorta

- **Valves** allow blood in one direction only. The tricuspid valve (three flaps) and bicuspid or mitral valve (two flaps) lie between atria and ventricles. **Semilunar valves** guard the exits of the pulmonary artery and the aorta.
- The **septum** separates the right half from the left, so oxygenated and deoxygenated blood never mix.
- The **left ventricle has the thickest wall** because it must push blood around the whole body at high pressure. The atria have thin walls since they only push blood into the ventricles.

Vessels entering and leaving the heart

VESSEL	DIRECTION	BLOOD
Superior vena cava, inferior vena cava	Into right atrium	Deoxygenated
Pulmonary artery	Out of right ventricle	Deoxygenated
Pulmonary veins (four)	Into left atrium	Oxygenated
Aorta	Out of left ventricle	Oxygenated

BOARD ALERT

Pulmonary artery and pulmonary vein are the two exceptions to the rule 'arteries carry oxygenated blood and veins carry deoxygenated'. The **pulmonary artery carries deoxygenated blood** and the **pulmonary vein carries oxygenated blood**.

Systole, diastole and the heartbeat

- **Systole** is the contraction of a chamber, **diastole** its relaxation.
- In one heartbeat (cardiac cycle, about 0.8 s at 72 beats per minute) the two atria contract first (atrial systole, about 0.1 s), then the ventricles (ventricular systole, about 0.3 s), then all chambers relax together (about 0.4 s).
- The characteristic 'lub-dub' sound is caused by the closure of valves. 'Lub' is the closing of the tricuspid and bicuspid valves at the start of ventricular systole. 'Dub' is the closing of the semilunar valves at the start of diastole.
- The beat starts in a patch of special muscle called the **sinoatrial (SA) node**, the natural pacemaker.

Blood Vessels



ARTERY
thick elastic muscular wall
narrow lumen, no valves

VEIN
thin wall, wide lumen
has valves

CAPILLARY
wall one cell thick
lumen = one RBC wide

FIG 8.3 CROSS SECTIONS OF BLOOD VESSELS (NOT TO SCALE)

FEATURE	ARTERY	VEIN	CAPILLARY
Direction	Away from the heart	Towards the heart	Between arteries and veins, in tissues
Wall	Thick, muscular, elastic	Thin, less muscle	One cell thick (endothelium)
Lumen	Narrow	Wide	Very narrow, as wide as one RBC
Valves	Absent (except at the heart base)	Present, to stop backflow	Absent
Pressure	High; blood flows in pulses	Low	Falling
Blood	Usually oxygenated	Usually deoxygenated	Exchange of gases, food and wastes

Structure and function: arteries have elastic walls to withstand pressure from the heart and to push the blood on; veins have valves because pressure is low, and the movement of skeletal muscles squeezes blood forward; capillaries have walls one cell thick so that oxygen and food can diffuse out and CO_2 and waste can diffuse in.

Circulation of Blood: Double Circulation

In humans the blood passes through the heart **twice** in one complete round of the body. This is called **double circulation**.

1. Pulmonary circulation
Right ventricle → pulmonary artery → lungs (CO2 out, O2 in) → pulmonary veins → left atrium.

2. Systemic circulation
Left ventricle → aorta → arteries → capillaries of the body → veins → vena cavae → right atrium.



KEY POINT

Advantage of double circulation: oxygenated and deoxygenated blood are kept apart, so tissues get fully oxygenated blood at high pressure, which supports the high metabolic rate of warm-blooded animals such as birds and mammals.

ORGAN	BLOOD VESSELS IN	BLOOD VESSELS OUT
Heart (muscle)	Coronary arteries	Cardiac veins into the right atrium
Liver	Hepatic artery (oxygenated) and hepatic portal vein (digested food)	Hepatic vein to inferior vena cava
Kidney	Renal artery from the aorta	Renal vein to inferior vena cava
Lungs	Pulmonary artery	Pulmonary veins

Pulse and blood pressure

- **Pulse:** the wave of expansion and recoil that passes along an artery each time the left ventricle contracts. It is felt at the wrist (radial artery) or the neck. Resting rate: about 72 per minute, the same as the heart rate.

- **Blood pressure:** the force of blood against the walls of the arteries. Normal is about **120/80 mm of mercury** (systolic over diastolic). It is measured with a **sphygmomanometer**. Persistently high pressure is called hypertension.

The Hepatic Portal System

Usually blood flows from the heart through capillaries and back to the heart. In a portal system, a vein starts as capillaries in one organ and ends as capillaries in another before blood returns to the heart. The **hepatic portal vein** collects blood from the stomach, intestine, spleen and pancreas and carries it to the liver, where it breaks up into capillaries (sinusoids).

- **Significance:** the liver receives the digested food **before** it reaches the rest of the body.
- Excess glucose is stored as glycogen, so that blood sugar does not rise suddenly. Glycogen is converted back to glucose when needed.
- Harmful substances and drugs absorbed from the gut are made less toxic (detoxification). Excess amino acids are broken down to urea.
- The liver also stores vitamins and iron, and makes bile and plasma proteins.

Blood Groups: ABO System and Rh Factor

The ABO groups were discovered by **Karl Landsteiner** in 1900. They depend on proteins called **antigens** on the surface of RBCs, and the **antibodies** in the plasma.

BLOOD GROUP	ANTIGEN ON RBC	ANTIBODY IN PLASMA	CAN DONATE TO	CAN RECEIVE FROM
A	A	Anti-B	A, AB	A, O
B	B	Anti-A	B, AB	B, O
AB	A and B	None	AB only	A, B, AB, O (**universal recipient**)
O	None	Anti-A and anti-B	A, B, AB, O (**universal donor**)	O only

If incompatible blood is transfused, the recipient's antibodies make the donor's red cells stick together (**agglutination**), which can block vessels and be fatal. This is why blood is always cross-matched.

Rh factor

People whose RBCs carry the **Rh antigen** (found first in rhesus monkeys) are **Rh positive**. Those without it are **Rh negative**. Rh negative people do not have anti-Rh antibodies naturally, but they make them if exposed to Rh positive blood.

KEY POINT

Rh incompatibility in pregnancy: if an Rh-negative mother carries an Rh-positive baby, some of the baby's blood may enter the mother's circulation at birth, so she makes anti-Rh antibodies. In the next pregnancy with an Rh-positive child, these antibodies cross the placenta and destroy the baby's RBCs (**erythroblastosis foetalis**). An injection of anti-D to the mother after the first delivery prevents this.

REAL LIFE

Blood banks store each group separately. In an emergency when the group is not known, O negative blood is given, since it has no A, B or Rh antigens.

HOT QUESTION

A man of blood group A marries a woman of blood group B. Their first child has blood group O. What are the genotypes of the parents, and which other blood groups could their children have?

HOT QUESTION

Why is the pressure in the pulmonary artery lower than in the aorta, and which part of the heart explains this?

Mind Map Recap

BRANCH	WHAT TO REMEMBER
Blood	Plasma (55%) + RBC + WBC + platelets
Clotting	Thromboplastin + Ca2+: prothrombin → thrombin → fibrinogen to fibrin
Heart	4 chambers; tricuspid and bicuspid valves; thick left ventricle; lub = AV valves closing
Vessels	Artery (thick wall), vein (valves), capillary (one cell thick)
Circulation	Double: pulmonary + systemic
Lymph	Tissue fluid in lymph vessels; spleen, tonsils, nodes
Groups	ABO antigens; O universal donor, AB universal recipient; Rh
Portal system	Gut → hepatic portal vein → liver → hepatic vein

Practice Questions

#	QUESTION	ANSWER
1	Why do mammalian red blood cells lack a nucleus?	To create space for more haemoglobin so they can carry more oxygen.
2	Name the pigment in RBCs and the metal in it.	Haemoglobin; iron.
3	Name the vessel that carries oxygenated blood from the lungs to the heart.	Pulmonary vein into the left atrium.
4	Why is the wall of the left ventricle thicker than the right?	It pumps blood to the whole body at high pressure, whereas the right ventricle pumps only to the nearby lungs.
5	What causes the 'lub' and 'dub' sounds of the heart?	Lub: closing of the tricuspid and bicuspid valves. Dub: closing of the semilunar valves.
6	Explain the role of platelets in blood clotting.	They release thromboplastin which, with calcium, begins the chain converting prothrombin to thrombin and fibrinogen to fibrin.
7	Differentiate between an artery and a vein (any three points).	Artery: carries blood away from the heart, thick walls, narrow lumen, no valves, high pressure. Vein: towards the heart, thin walls, wide lumen, valves, low pressure.
8	Why is blood group O called the universal donor?	Its RBCs have no A or B antigens, so no recipient antibodies can clump them.
9	What is the importance of the hepatic portal vein?	It carries digested food from the intestine straight to the liver, which regulates sugar, removes toxins and processes amino acids.
10	Name the two organs that are lymphatic organs and one function of each.	Spleen: destroys old RBCs. Tonsils: trap germs entering the throat.

09

CHAPTER · BIOLOGY

The Excretory System

Your cells make waste all day long, and a slow build-up would poison you. The excretory system filters the blood, discards what is harmful and keeps what is useful. The star of this chapter is the nephron.



IN THIS CHAPTER

- Excretory organs
- Urinary system
- Kidney structure
- Nephron
- Ultrafiltration
- Reabsorption and secretion
- Urine

What Is Excretion?

KEY POINT

Excretion is the removal of the harmful metabolic waste products from the body. (Removal of undigested food from the gut is **egestion**, not excretion.)

ORGAN	WHAT IT REMOVES
Kidneys	Urea, uric acid, creatinine, excess water and salts (as urine). The main excretory organs.
Skin	Sweat: water, salts and a little urea. Also cools the body.
Lungs	Carbon dioxide and water vapour.
Liver	Makes urea from excess amino acids; breaks down haemoglobin to bile pigments (bilirubin, biliverdin) that leave in the bile.
Large intestine	Some salts and bile pigments, with the faeces.

The Urinary System

PART	FUNCTION
Two **kidneys**	Filter blood and form urine
Two **ureters**	Muscular tubes that carry urine from kidneys to the bladder by peristalsis
Urinary bladder	Muscular bag that stores urine until it is passed out (micturition)
Urethra	Tube carrying urine out of the body

The Kidney

The kidneys are a pair of dark red, bean-shaped organs about 10 cm long, lying against the back wall of the abdomen on either side of the backbone, protected by the lower ribs. The right is slightly lower than the left because of the liver. The concave side faces inwards and has a notch, the **hilum**, where the renal artery and nerves enter and the renal vein and ureter leave.

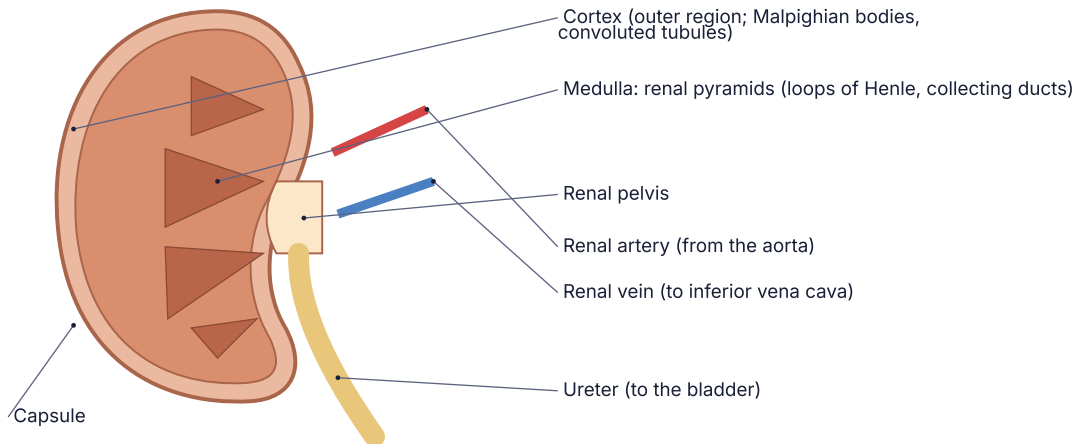


FIG 9.2 LONGITUDINAL SECTION OF A HUMAN KIDNEY (SCHEMATIC)

- The **renal artery** branches from the aorta and brings blood with urea. The **renal vein** carries filtered blood to the inferior vena cava.
- The **cortex** contains the Malpighian bodies and the convoluted tubules. The **medulla** contains the loops of Henle and the collecting ducts arranged in triangles called **renal pyramids**.
- The tips of the pyramids open into cup-shaped **calyces**, which lead to the **renal pelvis**, and this to the ureter.
- Each kidney contains roughly **one million nephrons**.

The Nephron: Working Unit of the Kidney

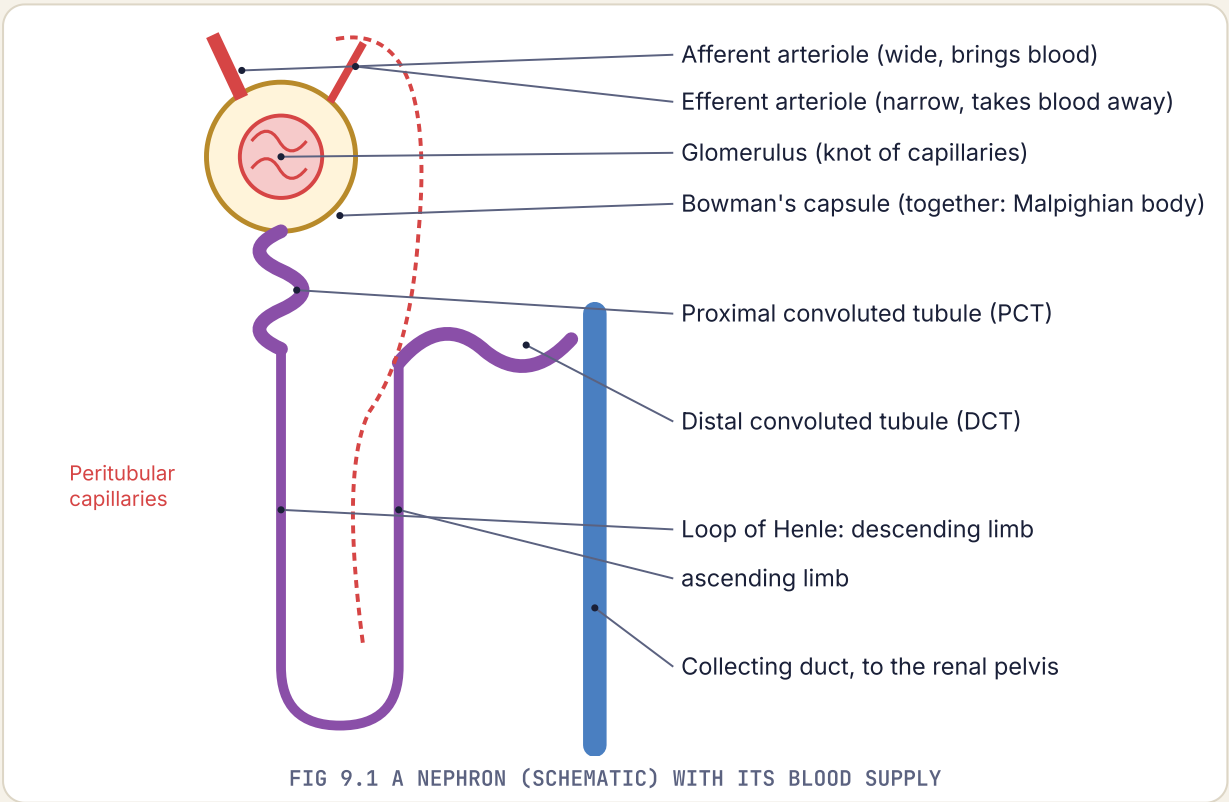


FIG 9.1 A NEPHRON (SCHEMATIC) WITH ITS BLOOD SUPPLY

PART	FUNCTION
Bowman's capsule + glomerulus (Malpighian body)	Ultrafiltration of blood
Proximal convoluted tubule	Selective reabsorption of glucose, amino acids, most of the water (about two-thirds) and salts
Loop of Henle	Reabsorption of water and salts; builds a salty region so that concentrated urine can be made
Distal convoluted tubule	Tubular secretion (H^+, K^+, creatinine, drugs); adjusts salts and pH; water reabsorption under ADH
Collecting duct	Final reabsorption of water under ADH; carries urine to the renal pelvis

Three Steps in the Formation of Urine

1. 1. Ultrafiltration
Blood enters the glomerulus through the wide afferent arteriole and leaves through the narrower efferent arteriole. The difference creates high pressure, forcing water, glucose, amino acids, urea and salts through the thin walls into Bowman's capsule. Blood cells and large proteins are too big and stay behind. The fluid collected is the glomerular filtrate. About 180 litres are filtered each day.

2. 2. Selective reabsorption
As the filtrate flows along the tubule, useful substances are taken back into the blood in the surrounding capillaries. All glucose and amino acids, most of the water and many salts return. This takes place mainly in the proximal tubule. Glucose is reabsorbed by active transport.

3. 3. Tubular secretion
Some wastes still in the blood, such as excess H^+ and K^+ ions, creatinine and certain drugs, are actively passed from capillaries into the tubule (mainly the distal tubule). This also helps adjust the blood pH.

The filtrate that remains, now only 1 to 1.5 litres a day, is **urine**. It flows from the collecting ducts to the pelvis, ureter and bladder.

SUBSTANCE	BLOOD PLASMA	GLOMERULAR FILTRATE	URINE
Proteins and blood cells	Present	Absent	Absent
Glucose	Present	Present	Absent (in a healthy person)
Urea	Little	Present	Much more
Water	About 90%	About 90%	About 95%

Urine is a pale yellow fluid (colour from the pigment **urochrome**) that is slightly acidic. It is about 95% water, 2.5% urea and 2.5% other substances such as salts, creatinine and uric acid.

SIDEKICK TIP

F-R-S: Filter, Reabsorb, Secrete. Filter everything small; reabsorb what the body wants; secrete what is left of the unwanted.

BOARD ALERT

Do not say the glomerulus filters out urea and keeps proteins because of 'selection'. Filtration is by **size** and pressure. The **selection** happens later during reabsorption.

When Kidneys Fail

- **Kidney stones** are hard deposits of calcium salts or uric acid that block the passage of urine.
- In kidney failure, wastes build up. **Dialysis** is an artificial method: the patient's blood is passed through a machine in which waste products diffuse out across a membrane into a dialysis fluid, and the cleaned blood is returned.
- A **kidney transplant** replaces a failed kidney with a healthy one from a donor. The donor must have a compatible tissue type so that the body does not reject it.

REAL LIFE

Diabetes can be first detected by testing urine for glucose. The presence of glucose in urine means that the amount filtered was more than the tubules can reabsorb.

HOT QUESTION

If the efferent arteriole were as wide as the afferent arteriole, what would happen to the formation of glomerular filtrate, and why?

HOT QUESTION

A person who drinks very little water passes a small volume of dark yellow urine. Explain how the kidney brings this about, naming the hormone involved.

Mind Map Recap

BRANCH	WHAT TO REMEMBER
Excretion	Kidneys, skin, lungs, liver
Urinary system	Kidneys → ureters → bladder → urethra
Kidney	Cortex, medulla (pyramids), pelvis; ~1 million nephrons each
Nephron	Malpighian body, PCT, loop of Henle, DCT, collecting duct
Urine formation	Ultrafiltration, selective reabsorption, tubular secretion
Urine	95% water, 2.5% urea, no glucose or proteins

Practice Questions

#	QUESTION	ANSWER
1	Name the parts of the urinary system in the order in which urine flows through them.	Kidney, ureter, urinary bladder, urethra.
2	What is a nephron? Name its parts.	The structural and functional unit of the kidney: Bowman's capsule with glomerulus, proximal convoluted tubule, loop of Henle, distal convoluted tubule and collecting duct.
3	Why is filtration in the glomerulus called ultrafiltration?	It occurs under high pressure through very fine pores, so only molecules smaller than proteins pass through.
4	Name two substances reabsorbed entirely under normal conditions.	Glucose and amino acids.
5	What is tubular secretion?	Movement of wastes such as H+, K+ and creatinine from blood into the nephron tubule.
6	Which blood vessels bring blood to and take blood from the kidney?	Renal artery and renal vein.
7	State two differences between glomerular filtrate and urine.	The filtrate has glucose and amino acids but urine does not; urine has a much higher concentration of urea.
8	What is the function of the loop of Henle?	To reabsorb water and salts and to create the concentration gradient that lets the kidney make concentrated urine.
9	Distinguish between excretion and egestion.	Excretion removes metabolic wastes from cells; egestion is the expulsion of undigested food.

10

CHAPTER · BIOLOGY

The Nervous System and Sense Organs

You touch a hot pan and your hand has moved before you feel the pain. How? This chapter traces the wiring of the body: the neurons, the brain and spinal cord, reflex arcs, and the eye and ear that feed the system with information.



IN THIS CHAPTER

- Neuron
- Central, peripheral and autonomic
- Brain
- Spinal cord
- Reflex action
- Eye
- Ear

The Neuron

The nervous system is made of cells called **neurons**. A neuron is a long cell built to carry electrical signals, called **nerve impulses**, quickly from one part of the body to another.

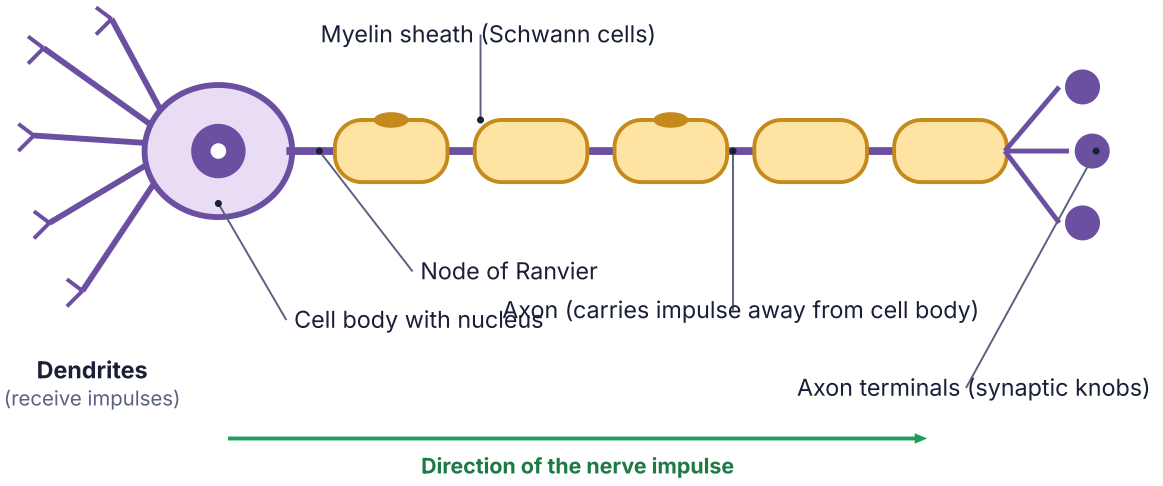


FIG 10.1 A MOTOR NEURON

PART	FUNCTION
Cell body (cyton)	Contains the nucleus and most of the cytoplasm; keeps the neuron alive
Dendrites	Short branched fibres that receive impulses from other neurons or from receptors and bring them to the cell body
Axon	A long fibre that carries impulses away from the cell body. It may be over a metre long (as in the neurons of the leg)
Myelin sheath	Fatty insulating covering around many axons, formed by Schwann cells. Makes conduction faster. The gaps between sections are the **nodes of Ranvier**
Axon terminals	The branched ends that pass the impulse across a gap, the **synapse**, to the next cell by releasing chemical messengers (neurotransmitters)

TYPE OF NEURON	ROLE
Sensory (afferent)	Carries impulses from receptors (sense organs) to the brain or spinal cord
Motor (efferent)	Carries impulses from the brain or spinal cord to effectors (muscles, glands)
Relay (association or intermediate)	Connects sensory and motor neurons inside the brain or spinal cord

SIDEKICK TIP

Dendrite = Door in; Axon = Away out. The impulse flows dendrite → cell body → axon → synapse, in that order, and only in one direction.

Divisions of the Nervous System

DIVISION	PARTS	NOTES
Central nervous system (CNS)	Brain and spinal cord	The control centre. Protected by the skull and the vertebral column, wrapped in three membranes (meninges) and bathed in cerebrospinal fluid that acts as a cushion
Peripheral nervous system (PNS)	12 pairs of cranial nerves and 31 pairs of spinal nerves	Connects the CNS to the rest of the body
Autonomic nervous system (ANS)	Sympathetic and parasympathetic nerves	Controls involuntary actions of internal organs, such as the heart, gut and glands. Part of the PNS

	SYMPATHETIC	PARASYMPATHETIC
Prepares body for	Emergency: 'fight or flight'	Rest and digestion: 'rest and digest'
Heart rate	Increases	Decreases
Pupil	Dilates	Constricts
Digestion	Slows	Speeds up
Breathing passages	Widened	Narrowed
Blood sugar	Rises	Not affected

The Brain

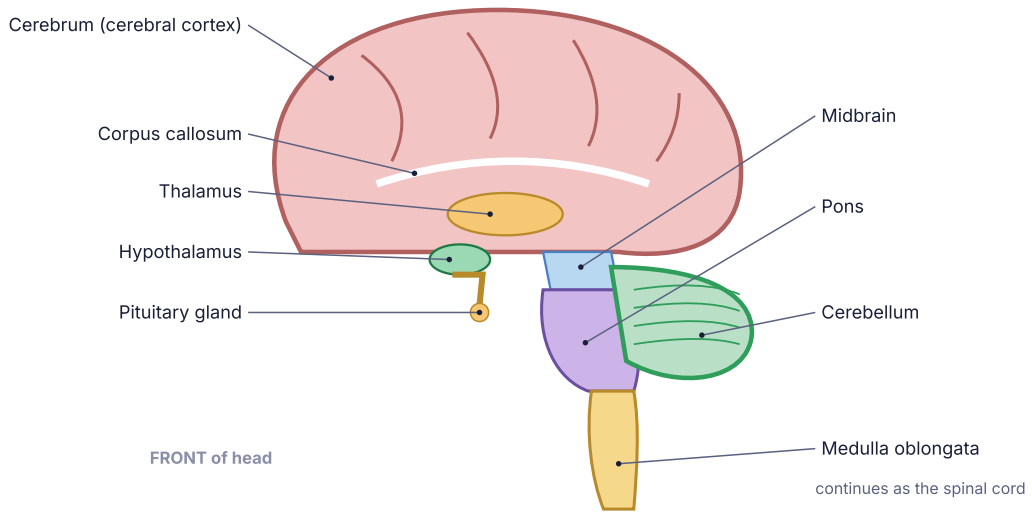


FIG 10.3 THE HUMAN BRAIN, SIDE VIEW IN SECTION (SCHEMATIC)

REGION	PARTS	FUNCTIONS
Forebrain	**Cerebrum**	The largest part. Two halves (cerebral hemispheres) joined by the corpus callosum. The folded outer layer (cortex) is grey matter. Seat of intelligence, thinking, memory, will power, speech, and conscious feeling from the senses; controls voluntary actions.
	Thalamus	Relay station: passes sensory signals to the cerebrum. Involved in pain and consciousness.
	Hypothalamus	Regulates body temperature, hunger, thirst and water balance; controls the pituitary gland; emotions such as anger and fear.
Midbrain		Controls reflex movements of the eyes and head in response to sight and sound.
Hindbrain	**Cerebellum**	Maintains balance and posture; coordinates smooth muscle movement (walking, riding a cycle). Does not start the movement.
	Pons	Connects the cerebrum and cerebellum and the two halves of the cerebellum; helps control breathing.
	Medulla oblongata	Controls involuntary actions: heartbeat, breathing, blood pressure, swallowing, coughing, sneezing, vomiting. Joins the spinal cord.



KEY POINT

Grey and white matter. Grey matter is cell bodies and is darker; white matter is nerve fibres wrapped in myelin. In the **brain**, grey matter is on the **outside** and white matter inside. In the **spinal cord** it is reversed: **white outside, grey inside** (butterfly or H shape).

The spinal cord

The spinal cord is a cylinder of nervous tissue about 45 cm long, continuing from the medulla through the vertebral column. A tiny **central canal** filled with fluid runs through it. It has two jobs: to carry impulses up to the brain and down from it, and to control reflex actions. Each spinal nerve joins it by two roots: the **dorsal root** (sensory, with a swelling called the ganglion) and the **ventral root** (motor).



BOARD ALERT

Do not mix up cerebrum and cerebellum. **Cerebrum** = thinking and voluntary action. **Cerebellum** = balance and coordination. A drunk person staggers because alcohol affects the cerebellum.

Voluntary, Involuntary and Reflex Actions

TYPE	MEANING	CONTROLLED BY	EXAMPLES
Voluntary	Done on purpose, with thought	Cerebrum	Writing, kicking a ball, talking
Involuntary	Done automatically, without thinking	Medulla and other lower parts, through the autonomic system	Heartbeat, breathing, peristalsis, sweating
Reflex	Sudden, automatic response to a stimulus that does not wait for the brain	Spinal cord (or brain stem for cranial reflexes)	Pulling the hand from a hot object, knee jerk, blinking, sneezing

The reflex arc

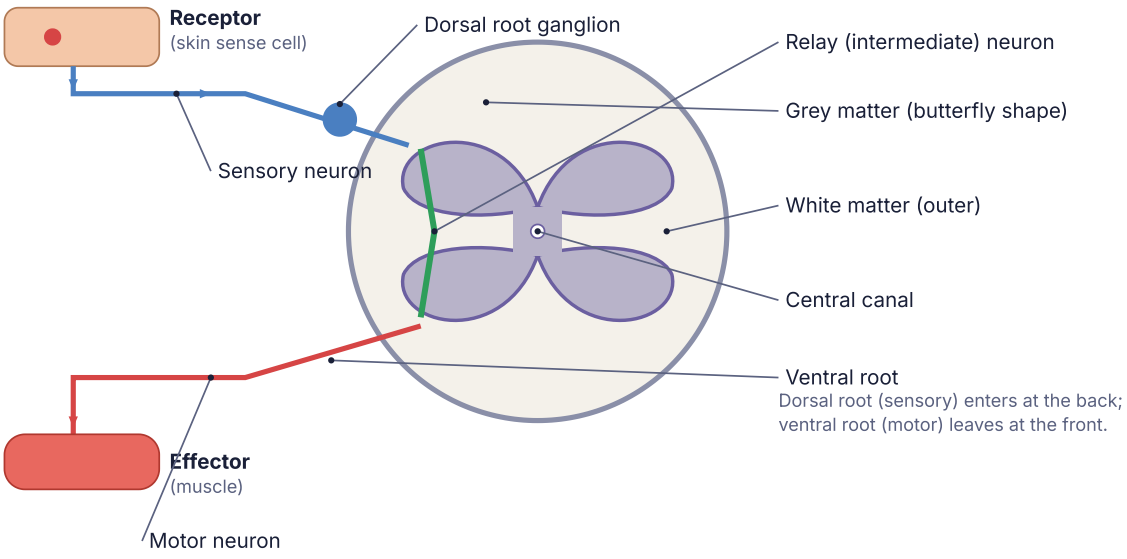


FIG 10.2 THE REFLEX ARC (SPINAL CORD IN CROSS SECTION)

A **reflex arc** is the pathway taken by the impulse in a reflex action:

Stimulus → Receptor → Sensory neuron → Spinal cord (relay neuron) → Motor neuron → Effector → Response

The brain is told about it only after the muscle has already acted, which is why the reflex is so fast. The shortcut protects the body from harm.

	NATURAL (INBORN) REFLEX	ACQUIRED (CONDITIONED) REFLEX
Origin	Present from birth	Developed by learning and experience
Inheritance	Inherited	Not inherited
Brain involved?	No, spinal cord is enough	Yes, cerebrum
Permanence	Permanent	May fade if not reinforced
Examples	Knee jerk, blinking, sucking by a baby, withdrawal from pain	A dog salivating at the sound of a bell (Pavlov); stopping at a red light; mouth watering at the sight of food



FUN FACT

Ivan Pavlov rang a bell every time he fed a dog. After some time the dog salivated at the bell alone, even with no food. That is a conditioned reflex.

The Eye

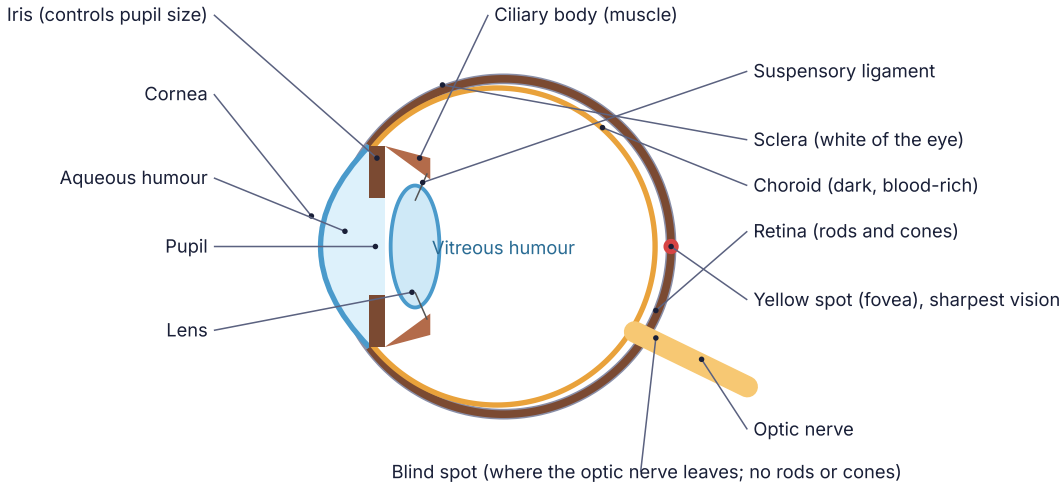


FIG 10.4 THE HUMAN EYE IN SECTION (SCHEMATIC)

PART	FUNCTION
Sclera	Tough white outer coat that protects and keeps the shape
Cornea	Clear front window; does most of the bending (refraction) of light
Choroid	Dark layer rich in blood vessels; nourishes the eye and prevents internal reflection
Iris	Coloured muscular diaphragm; adjusts the size of the pupil
Pupil	Central opening for the entry of light
Lens	Transparent, flexible and convex; focuses light on the retina
Ciliary muscles and suspensory ligaments	Change the shape of the lens
Aqueous humour	Watery fluid between cornea and lens; keeps the cornea in shape, supplies food
Vitreous humour	Jelly-like fluid behind the lens; keeps the eyeball round
Retina	Inner light-sensitive layer. **Rods**: dim light, black and white. **Cones**: bright light and colour (red, green, blue)
Yellow spot (fovea)	Point of sharpest vision; many cones
Blind spot	No rods and cones, where the optic nerve leaves. No image is seen here
Optic nerve	Carries impulses to the brain (occipital lobe of the cerebrum)

Path of light: cornea → aqueous humour → pupil → lens → vitreous humour → retina. The image formed on the retina is real and **inverted**. The brain turns it the right way up.

Accommodation

Accommodation is the ability of the eye to focus on objects at different distances by changing the curvature of the lens.

	NEAR OBJECT	DISTANT OBJECT
Ciliary muscles	Contract	Relax
Suspensory ligaments	Slacken	Tighten
Lens	Thicker, more convex	Thinner, flatter
Focal length	Shorter	Longer

Adaptation, and stereoscopic vision

- **Adaptation to light:** in bright light the circular muscles of the iris contract and the pupil narrows; in dim light the radial muscles contract and the pupil widens. After coming in from sunlight, you cannot see in a dark

room for a few moments. The rods slowly regain sensitivity (dark adaptation). Vitamin A is needed to make the pigment in the rods; its lack causes night blindness.

- **Stereoscopic (binocular) vision:** the two eyes see an object from slightly different angles. The brain combines the two views into one picture with depth, giving a three-dimensional impression and the ability to judge distance.

Defects of vision

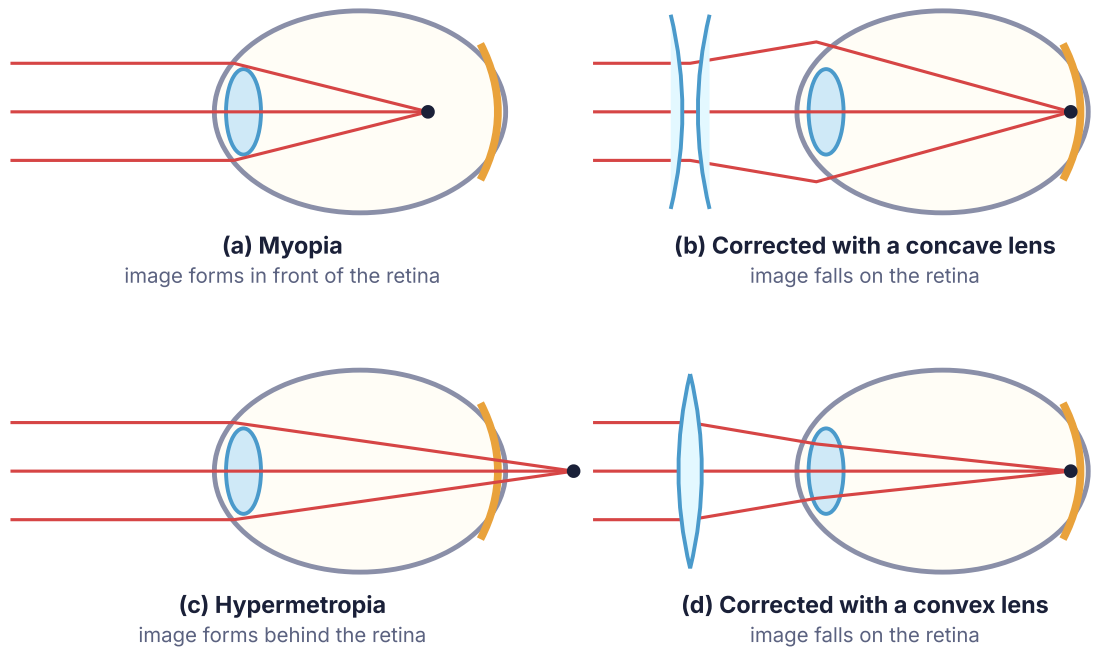


FIG 10.5 DEFECTS OF VISION AND THEIR CORRECTION (SCHEMATIC)

DEFECT	CAUSE	EFFECT	CORRECTION
Myopia (short-sightedness)	Eyeball too long or lens too curved	Near objects are clear; distant objects blurred. Image falls in front of the retina	Concave (diverging) lens
Hypermetropia (long-sightedness)	Eyeball too short or lens too flat	Distant objects are clear; near objects blurred. Image falls behind the retina	Convex (converging) lens
Presbyopia	Loss of elasticity of the lens and weakening of ciliary muscles with age	Difficulty in seeing near objects	Bifocal lenses (concave on top, convex below)
Astigmatism	Unequal curvature of the cornea or lens	Blurred or distorted vision in some directions	Cylindrical lens
Cataract	Lens becomes cloudy (opaque)	Gradual loss of vision	Surgery: the cloudy lens is replaced by an artificial lens

The Ear

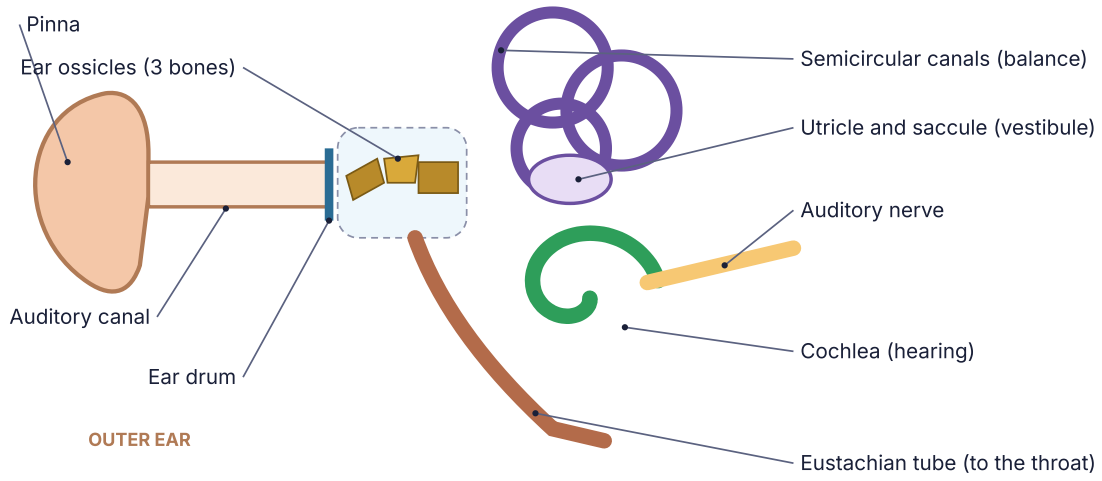


FIG 10.6 THE HUMAN EAR (SCHEMATIC)

PART	STRUCTURE	FUNCTION
Outer ear	Pinna, auditory canal, tympanic membrane (eardrum)	Collects sound waves and directs them to the eardrum, which vibrates. Wax and hairs in the canal trap dust.
Middle ear	Air-filled cavity with three tiny bones (malleus, incus, stapes), the oval window and the Eustachian tube	The ossicles carry and amplify the vibrations. The Eustachian tube links the cavity to the throat and equalises air pressure on both sides of the eardrum.
Inner ear	Cochlea (a coiled tube filled with fluid and lined with hair cells), the vestibule (utricle and saccule) and three semicircular canals	Cochlea: converts vibrations to nerve impulses (hearing). Semicircular canals and vestibule: balance.

Course of perception of sound

1. Collect
Sound waves are gathered by the pinna and travel down the auditory canal.

2. Vibrate
They make the eardrum vibrate.

3. Amplify
The vibrations are carried across the middle ear by the malleus, incus and stapes. These act as a lever system and make them stronger.

4. Transfer
The stapes pushes on the oval window, which sets up waves in the fluid of the cochlea.

5. Convert
The waves bend the hair cells of the organ of Corti in the cochlea, which produce nerve impulses.

6. Perceive
The auditory nerve carries the impulses to the temporal lobe of the cerebrum, where they are heard as sound.

The ear and balance

The three **semicircular canals** lie at right angles to one another and contain fluid. When the head rotates, the fluid moves and bends hair cells, so the brain senses the rotation. The **utricle and saccule** contain tiny crystals that press on hair cells in different ways when the head is tilted and tell the body its position relative to gravity. The cerebellum uses this information to keep the body balanced.

REAL LIFE

Pressure change when an aircraft takes off or lands, or when you dive, pushes the eardrum. Swallowing or yawning opens the Eustachian tube and equalises the pressure, so the ear 'pops'.

HOT QUESTION

A boy is able to read the board but not the notebook in his hand, while his grandfather can read the newspaper only when he holds it at arm's length. Name the defect in each case, say what is wrong with the eye, and name the lens needed.

HOT QUESTION

Why can a cricketer catch a fast ball more easily using both eyes than with one eye closed?

Mind Map Recap

BRANCH	WHAT TO REMEMBER
Neuron	Dendrite → cell body → axon → synapse; myelin; nodes of Ranvier
Divisions	CNS (brain + cord), PNS (cranial 12, spinal 31 pairs), ANS (sympathetic, parasympathetic)
Brain	Cerebrum (thinking), cerebellum (balance), medulla (involuntary), pons, thalamus, hypothalamus
Reflex	Receptor → sensory → relay → motor → effector
Eye	Light path; rods/cones; accommodation; myopia (concave), hypermetropia (convex)
Ear	Outer, middle (ossicles), inner (cochlea, canals); hearing and balance

Practice Questions

#	QUESTION	ANSWER
1	Name the parts of a neuron.	Cell body, dendrites, axon (with myelin sheath and nodes of Ranvier) and axon terminals.
2	Which part of the brain controls (a) balance, (b) heartbeat, (c) body temperature?	(a) Cerebellum (b) Medulla oblongata (c) Hypothalamus.
3	State two differences between a reflex action and a voluntary action.	A reflex is quick and automatic, controlled by the spinal cord. A voluntary action is deliberate and slower, controlled by the cerebrum.
4	Write the sequence of the reflex arc.	Stimulus, receptor, sensory neuron, spinal cord, motor neuron, effector, response.
5	Differentiate between natural and acquired reflexes with one example each.	Natural: inborn, e.g. knee jerk. Acquired: learned, e.g. salivation at a bell in a trained dog.
6	What are rods and cones?	Light-sensitive cells of the retina: rods work in dim light; cones work in bright light and detect colours.
7	How does the eye focus on a nearby object?	Ciliary muscles contract, suspensory ligaments slacken and the lens becomes thicker and more convex.
8	Name the defect in which the image forms in front of the retina and the lens used to correct it.	Myopia; concave lens.
9	What is the function of the Eustachian tube?	Equalises air pressure on the two sides of the eardrum.
10	Which parts of the inner ear help in (a) hearing (b) balance?	(a) Cochlea (b) Semicircular canals, utricle and saccule.

11

CHAPTER · BIOLOGY

The Endocrine System

Nerves send quick text messages. Hormones send letters that take longer to arrive but last much longer. This chapter introduces the glands that make them, what each hormone does, and what goes wrong when there is too little or too much.



IN THIS CHAPTER

- Endocrine vs exocrine
- Pituitary
- Thyroid
- Adrenal
- Pancreas
- Feedback with TSH

Glands and Hormones

A **gland** is an organ that makes and releases a useful substance. **Exocrine glands** release their product through a duct. **Endocrine glands** are ductless: they release their products, **hormones**, directly into the blood.

KEY POINT

Hormone: a chemical messenger secreted in very small amounts by an endocrine gland, carried by the blood to a target organ, where it controls a particular activity.

FEATURE	ENDOCRINE GLAND	EXOCRINE GLAND
Duct	Absent (ductless)	Present
Secretion goes into	Blood	Duct leading to a surface or cavity
Product	Hormones	Enzymes, sweat, saliva, tears, milk, mucus
Examples	Pituitary, thyroid, adrenal	Salivary glands, sweat glands, gastric glands, liver
Mixed gland	The **pancreas** is both: islets of Langerhans (endocrine) and pancreatic juice through the duct (exocrine). Gonads (testes and ovaries) also make gametes and hormones.	

NERVOUS COORDINATION	HORMONAL COORDINATION
Electrical impulses along neurons	Chemicals carried by blood
Very fast	Slower
Short-lived effect	Long-lasting effect
Acts on a specific, local target	Can act on many organs

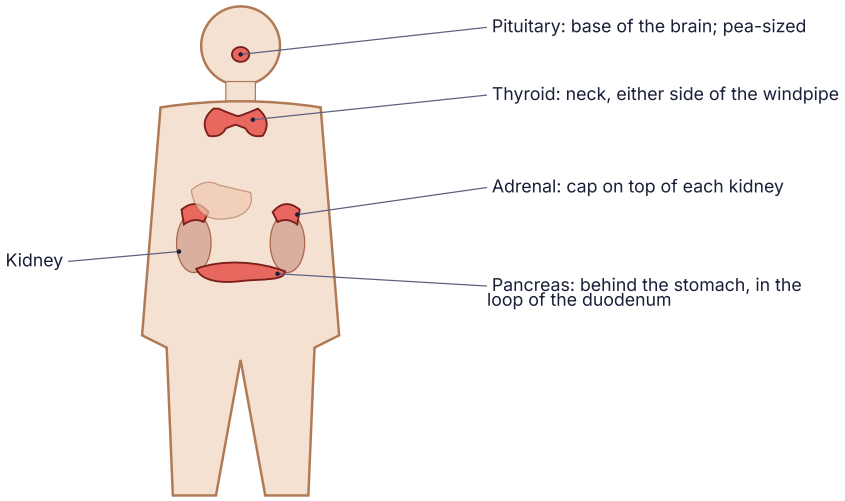


FIG 11.1 POSITION OF THE ENDOCRINE GLANDS STUDIED IN THE SYLLABUS (SCHEMATIC)

Pituitary Gland: The 'Master Gland'

The pituitary is a pea-sized gland that hangs from the hypothalamus at the base of the brain, protected in a small hollow of the skull bone. It is called the master gland because its hormones control other endocrine glands. The pituitary itself is controlled by the hypothalamus.

PART	HORMONE	ACTION	HYPOSECRETION	HYPERSECRETION
Anterior lobe	**Growth hormone (GH)**	Growth of bones and tissues	**Dwarfism** in children	**Gigantism** in children; **acromegaly** (enlarged face, hands, feet) in adults
	Tropic hormones: TSH, ACTH, FSH, LH	Stimulate other endocrine glands. TSH: thyroid; ACTH: adrenal cortex; FSH and LH: gonads	Underactivity of the target glands	Overactivity of the target glands
Posterior lobe	**ADH (antidiuretic hormone)**	Makes the kidney tubules reabsorb more water, so less urine is formed	**Diabetes insipidus:** very large amount of dilute urine, constant thirst	
	Oxytocin	Contraction of the uterus in childbirth; release of milk	Difficult labour	

SIDEKICK TIP

ADH = Anti-Diuretic: against making lots of urine. Sweat a lot, ADH rises, urine turns scanty and dark. Drink a lot, ADH falls, urine turns plentiful and pale.

Thyroid Gland

A butterfly-shaped gland in the neck, in front of the windpipe, just below the larynx. It makes **thyroxine**, which needs **iodine**. Thyroxine controls the body's **metabolic rate** (BMR), growth and development.

CONDITION	CAUSE	EFFECTS
Simple (iodine-deficiency) goitre	Too little iodine in the diet	Swelling of the thyroid in the neck
Cretinism	Hyposecretion in childhood	Stunted growth, poor mental development
Myxoedema	Hyposecretion in adults	Slow metabolism, weight gain, tiredness, dry skin, slow heartbeat
Hyperthyroidism (e.g. exophthalmic goitre)	Hypersecretion	Weight loss, fast heartbeat, sweating, nervousness, bulging eyes



REAL LIFE

Iodised salt prevents goitre. Iodine is naturally low in soil and water in hilly and mountainous regions, so people there are more likely to suffer from it.

Adrenal Glands

A pair of cap-shaped glands, one above each kidney. Each has an outer **cortex** and an inner **medulla**.

PART	HORMONES	ACTION	DISORDER
Cortex	**Cortical hormones**: glucocorticoids (cortisol: carbohydrate metabolism and stress), mineralocorticoids (aldosterone: reabsorption of Na^+ and loss of K^+ in the kidney) and small amounts of sex hormones	Control of blood sugar, salt and water balance	Hyposecretion: **Addison's disease** (weakness, low blood pressure). Hypersecretion: **Cushing's syndrome** (rounded face, high blood sugar and blood pressure)
Medulla	**Adrenaline** (and noradrenaline)	The 'emergency hormone': heart rate and breathing speed up, blood sugar rises, pupils dilate, more blood to the muscles, skin pales	Hypersecretion: anxiety, high blood pressure



KEY POINT

Adrenaline is the **hormone of fight, flight and fright**. It prepares the body for emergency action within seconds, and works along with the sympathetic nervous system.

Pancreas

A leaf-shaped gland behind the stomach, lying in the loop of the duodenum. Clusters of cells called the **islets of Langerhans** make two opposing hormones.

HORMONE	CELLS	ACTION
Insulin	Beta cells	Lowers blood sugar: helps cells take up glucose, and converts excess glucose into glycogen in the liver and muscle
Glucagon	Alpha cells	Raises blood sugar: converts glycogen in the liver back into glucose

CONDITION	CAUSE	SYMPTOMS
Diabetes mellitus	Hyposecretion of insulin (or the body not responding to it)	High blood sugar, glucose in urine, frequent urination, thirst, weight loss, tiredness
Hypoglycaemia	Hypersecretion of insulin, or too much insulin injected	Low blood sugar: sweating, trembling, dizziness, fainting

BOARD ALERT

Do not confuse **diabetes mellitus** (insulin, sugar in the urine) with **diabetes insipidus** (ADH, dilute urine, no sugar). Both cause lots of urine, but the cause and the hormone are different.

Feedback Control: The TSH Example

Hormone levels are kept steady by **negative feedback**: a rise in the final product switches off the signal that produced it.

1. Low thyroxine in blood
The hypothalamus detects this and stimulates the pituitary.

2. Pituitary releases TSH
TSH (thyroid-stimulating hormone) travels in the blood to the thyroid.

3. Thyroid releases thyroxine
The level of thyroxine in the blood rises.

4. Feedback
High thyroxine tells the hypothalamus and pituitary to reduce TSH. The thyroid slows down. The level settles.

SIDEKICK TIP

The same principle works for blood sugar: rising glucose releases insulin, and falling glucose releases glucagon. Think of a thermostat that switches the heater on and off.

FUN FACT

The pituitary of an adult is only about the size of a pea (around 0.5 g) and yet controls the thyroid, adrenals, gonads and growth.

HOT QUESTION

A patient has a swollen neck, is gaining weight and feels tired and cold. Blood tests show a high TSH and a low thyroxine level. Which gland is faulty, what is the likely cause and what will happen to TSH if the patient is given thyroxine tablets?

HOT QUESTION

A student who is about to take a stage test notices a pounding heart, wide pupils and a dry mouth. Name the hormone responsible, the gland that releases it, and the part of the nervous system that works with it.

Mind Map Recap

BRANCH	WHAT TO REMEMBER
Endocrine vs exocrine	Ductless, hormones in blood vs ducts
Pituitary	GH, tropic hormones, ADH, oxytocin; master gland
Thyroid	Thyroxine; goitre, cretinism, myxoedema, hyperthyroidism
Adrenal	Cortex: cortical hormones; medulla: adrenaline
Pancreas	Insulin (down), glucagon (up); diabetes
Feedback	Thyroxine high → TSH low

Practice Questions

#	QUESTION	ANSWER
1	What are endocrine glands? Give two examples.	Ductless glands that release hormones into the blood, e.g. pituitary and thyroid.
2	Why is the pituitary called the master gland?	Its tropic hormones control the activity of other endocrine glands.
3	Name the hormone that (a) lowers blood sugar (b) helps the kidney conserve water.	(a) Insulin (b) ADH.
4	What is the effect of hyposecretion of growth hormone in childhood?	Dwarfism.
5	Which mineral is needed to make thyroxine? What disease results from its shortage?	Iodine; simple goitre.
6	State one function each of oxytocin and adrenaline.	Oxytocin: contraction of the uterus at birth. Adrenaline: prepares the body for fight or flight.
7	How do insulin and glucagon work together?	Insulin lowers blood sugar and glucagon raises it, so the sugar level remains steady.
8	Explain negative feedback using TSH.	When thyroxine rises, it suppresses TSH release from the pituitary, which reduces thyroid activity and lowers the thyroxine level.
9	Differentiate between an endocrine and an exocrine gland (any two points).	Endocrine: ductless and secretes hormones into blood. Exocrine: has ducts and secretes enzymes, sweat, etc.

12

CHAPTER · BIOLOGY

The Reproductive System

A single sperm meets a single egg, and nine months later there is a new person. This chapter follows the whole story: the organs, the gametes, fertilisation, the life support of the placenta, birth, the monthly cycle and the hormones behind it all.



IN THIS CHAPTER

- Male organs
- Female organs
- Sperm and ovum
- Fertilisation to birth
- Placenta
- Menstrual cycle
- Sex hormones
- Twins

Male Reproductive System

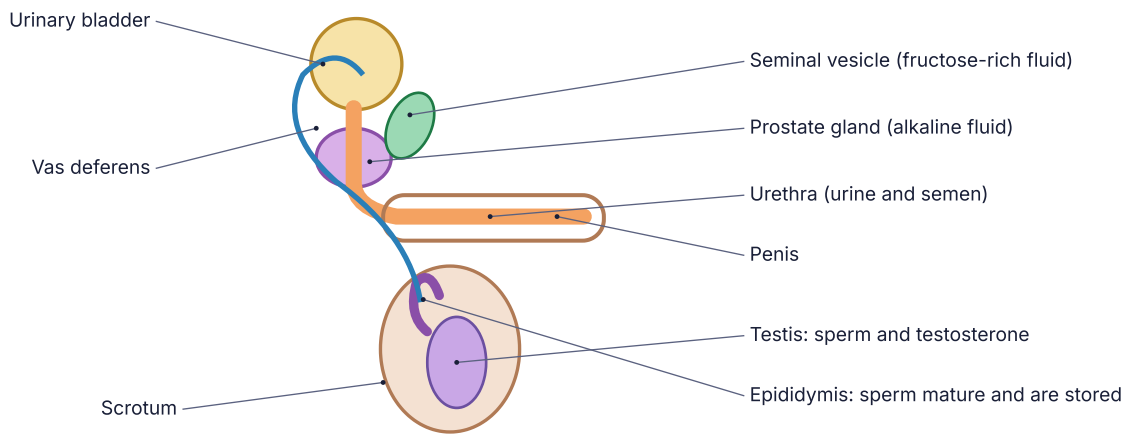


FIG 12.2 MALE REPRODUCTIVE SYSTEM (SIDE VIEW, SCHEMATIC)

ORGAN	FUNCTION
Testes (pair, in the scrotum)	Make sperm in the seminiferous tubules; the interstitial (Leydig) cells make testosterone
Scrotum	Pouch holding the testes outside the body, 2 to 3 degrees Celsius cooler, which suits sperm production
Epididymis	Coiled tube on each testis where sperm mature and are stored
Vas deferens	Carries sperm from the epididymis towards the urethra
Seminal vesicles	Add a fluid rich in fructose (food for sperm) and other substances
Prostate gland	Adds a milky, alkaline fluid that helps sperm movement and neutralises the acid in the vagina
Cowper's (bulbourethral) glands	Add a clear mucus that lubricates the urethra
Urethra and penis	The urethra carries both urine and semen. The penis places semen in the vagina

Semen is the mixture of sperm and the secretions of the seminal vesicles, prostate and Cowper's glands. The three glands together are the **male accessory glands**.

Female Reproductive System

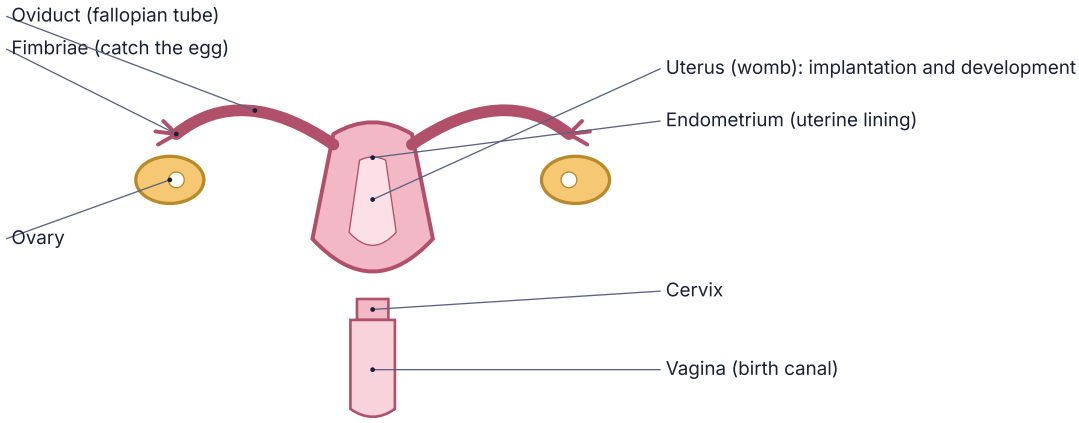


FIG 12.1 FEMALE REPRODUCTIVE SYSTEM (FRONT VIEW, SCHEMATIC)

ORGAN	FUNCTION
Ovaries (pair)	Produce ova (eggs) and the hormones oestrogen and progesterone
Oviducts (fallopian tubes)	Funnel-shaped ends with finger-like fimbriae catch the egg. The egg is carried by cilia and muscle movement. **Fertilisation occurs here.**
Uterus	Muscular, pear-shaped organ with a lining (endometrium). The embryo implants and grows here.
Cervix	Narrow neck of the uterus that opens into the vagina
Vagina	Receives the sperm; serves as the birth canal

Secondary Sexual Characters

At puberty (roughly 10 to 14 years in girls and 12 to 16 in boys), the sex hormones cause features that distinguish the sexes but are not directly part of the reproductive organs.

BOYS (TESTOSTERONE)	GIRLS (OESTROGEN)
Beard, moustache, hair on chest	Development of breasts
Deepening of the voice	Widening of the hips
Broad shoulders, more muscle	Start of menstruation
Hair in underarm and pubic regions	Hair in underarm and pubic regions

The Gametes

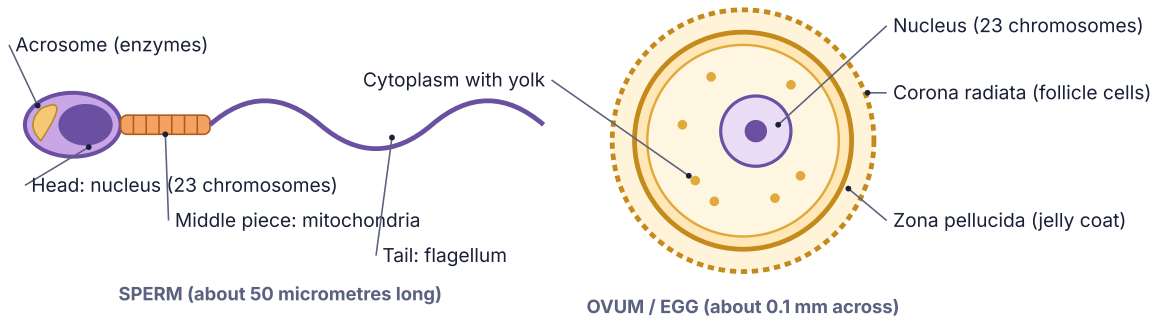


FIG 12.3 HUMAN SPERM AND OVUM (NOT TO THE SAME SCALE)

	SPERM	OVUM
Size	Very small (about 50 micrometres)	Largest human cell (about 0.1 mm)
Movement	Motile (tail)	Non-motile
Number	Millions in each ejaculation	One released a month (about 400 in a lifetime)
Food reserve	Almost none	Some yolk
Chromosome	23 (X or Y)	23 (always X)
Made in	Testes, from puberty throughout life	Ovaries, started before birth; released from puberty to menopause

From Fertilisation to Birth

TERM	MEANING
Fertilisation	Fusion of the nucleus of a sperm with that of an ovum to form a **zygote** (46 chromosomes). It occurs in the oviduct. Enzymes in the acrosome help the sperm to digest through the egg coverings. After one sperm has entered, the egg changes its membrane so that no other can.
Cleavage	The zygote divides repeatedly by mitosis as it travels to the uterus, forming a ball of cells (embryo)
Implantation	The embedding of the young embryo into the thick, soft endometrium of the uterus, about 6 to 7 days after fertilisation
Gestation	The period of development inside the uterus, about **9 months** (around 280 days from the last menstrual period)
Parturition	The process of giving birth. Oxytocin makes the uterus contract in strong waves (labour) to push out the baby

The Placenta and Foetal Membranes

After implantation, finger-like **villi** from the embryo grow into the uterine wall and form a disc-shaped organ, the **placenta**. The baby is attached to it by the **umbilical cord**, which carries two umbilical arteries (deoxygenated blood to the placenta) and one umbilical vein (oxygenated blood and nutrients to the baby). The mother's blood and the baby's blood do **not mix**; exchange takes place across thin walls.

ROLE OF THE PLACENTA	DETAILS
Nutrition	Passes glucose, amino acids, vitamins, minerals and antibodies from mother to foetus
Respiration	Oxygen goes in; carbon dioxide comes out by diffusion
Excretion	Urea and other wastes pass from foetus to mother's blood
Endocrine function	Makes hormones including hCG, oestrogen and progesterone, which maintain pregnancy
Protection	Acts as a barrier against many germs (but not all: some viruses and harmful substances, including alcohol and drugs, can cross)

Foetal membranes and amniotic fluid

- The **amnion** is a thin, tough sac around the foetus filled with **amniotic fluid**. The outer membrane, the **chorion**, forms the foetal part of the placenta. The allantois and yolk sac are small in humans.
- The fluid acts as a **shock absorber**, protects against injury, keeps a steady temperature, lets the foetus move freely and prevents it from sticking to the membranes. Just before birth, the amniotic sac breaks and the fluid is released ('the waters break').

The Menstrual Cycle

From puberty until about 45 to 50 years of age, the female reproductive system goes through a cycle of about **28 days**, which prepares the uterus for a possible pregnancy. The first cycle is the **menarche**, and the permanent end is the **menopause**.

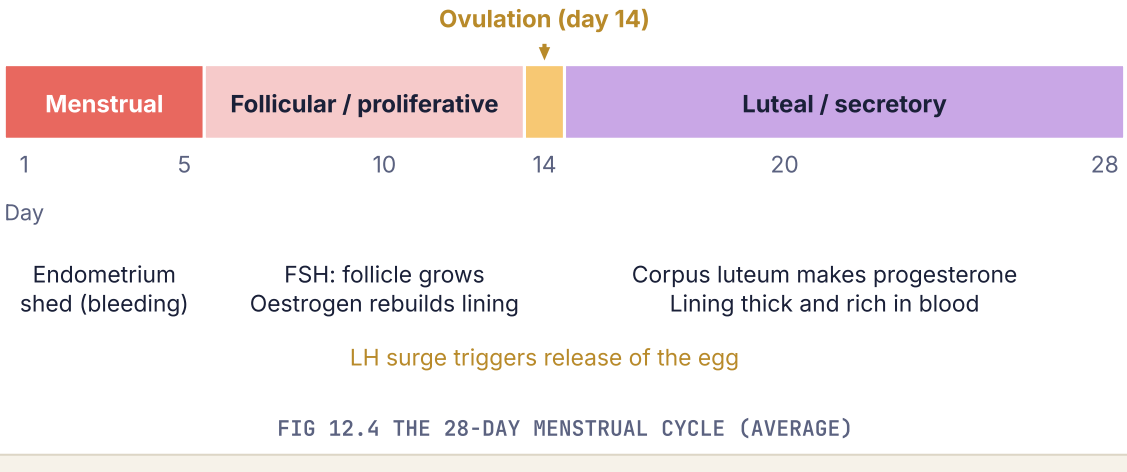


FIG 12.4 THE 28-DAY MENSTRUAL CYCLE (AVERAGE)

PHASE	DAYS	WHAT HAPPENS
Menstrual	1 to 5	The unused, thickened endometrium and blood leave through the vagina
Follicular (proliferative)	6 to 13	FSH from the pituitary makes a follicle in the ovary grow. The follicle releases oestrogen, which rebuilds the uterine lining
Ovulation	About day 14	A surge of LH makes the ripe follicle release an egg from the ovary
Luteal (secretory)	15 to 28	The empty follicle becomes the **corpus luteum** and makes progesterone, which thickens and enriches the lining. If the egg is not fertilised, the corpus luteum shrinks, hormone levels fall and menstruation begins again



KEY POINT

If fertilisation **does** occur, the corpus luteum continues to function (maintained by hCG), progesterone stays high, the lining remains, and **menstruation does not occur**. A missed period is one of the first signs of pregnancy.

Sex Hormones

HORMONE	MADE BY	FUNCTIONS
Testosterone	Testes (Leydig cells)	Sperm production; male secondary sexual characters
Oestrogen	Ovarian follicles (and placenta)	Female secondary sexual characters; repair and thickening of the uterine lining; development of the breasts
Progesterone	Corpus luteum and placenta	Prepares and maintains the uterine lining for implantation and pregnancy; prevents the uterus from contracting; helps the breasts get ready to make milk

Twins

	IDENTICAL TWINS	FRATERNAL TWINS
Origin	One zygote splits into two early in development	Two different eggs are released and fertilised by two different sperms
Scientific name	Monozygotic	Dizygotic
Sex	Always the same	Same or different
Genes	Identical	As different as ordinary brothers and sisters
Placenta	Often shared	Usually separate

FUN FACT

The ovum is the largest cell in the human body, yet it is only just visible to the naked eye as a tiny speck. The sperm is among the smallest.

REAL LIFE

Pregnant women are advised not to smoke or drink alcohol. Many harmful substances can pass across the placenta and affect the developing baby.

BOARD ALERT

Fertilisation takes place in the **oviduct**, not in the uterus. Implantation takes place in the uterus. These two words are often interchanged in exam answers.

HOT QUESTION

A woman's menstrual cycle is regular, 28 days. The first day of her last period was on the 1st of the month. When will ovulation take place and during which days is she most likely to conceive?

HOT QUESTION

Why do identical twins always have the same sex, whereas fraternal twins need not?

Mind Map Recap

BRANCH	WHAT TO REMEMBER
Male	Testis, epididymis, vas deferens, seminal vesicle, prostate, urethra
Female	Ovary, oviduct (fertilisation), uterus (implantation), cervix, vagina
Gametes	Sperm: acrosome, nucleus, mitochondria, tail. Ovum: large, yolk
Sequence	Fertilisation → zygote → implantation → placenta → gestation (9 months) → parturition
Placenta	Nutrition, respiration, excretion, hormones
Cycle	Menstruation 1-5, follicular, ovulation 14, luteal
Hormones	Testosterone, oestrogen, progesterone
Twins	Identical (one zygote) vs fraternal (two)

Practice Questions

#	QUESTION	ANSWER
1	What is the function of the acrosome of a sperm?	It contains enzymes that dissolve the egg coverings so that the sperm can enter.
2	Why are the testes located outside the body cavity?	Sperm need a temperature 2 to 3 degrees lower than body temperature to develop.
3	Where do fertilisation and implantation take place?	Fertilisation in the oviduct; implantation in the uterus.
4	Name the three functions of the placenta related to the embryo's life.	Nutrition, respiration and excretion (it also secretes hormones).
5	State two functions of amniotic fluid.	Shock absorber and protection from injury, and maintaining a constant temperature.
6	What happens to the endometrium during menstruation?	The unused lining breaks down and is shed with some blood through the vagina.
7	Which hormone is secreted by the corpus luteum? State its function.	Progesterone: maintains and prepares the uterine lining for pregnancy.
8	Differentiate between identical and fraternal twins.	Identical twins come from one zygote and are genetically alike and of the same sex; fraternal twins come from two zygotes and are genetically different.
9	Define gestation and parturition.	Gestation is the period of development in the uterus (about 9 months); parturition is the act of giving birth.
10	Name two secondary sexual characters in boys and in girls.	Boys: beard and deep voice. Girls: breast development and broad hips.

13

CHAPTER · BIOLOGY

Population Explosion

The world added its first billion people over all of human history, and its latest billion in barely a dozen years. This chapter looks at why the numbers grew so fast, what that does to people and resources, and how it can be slowed.



IN THIS CHAPTER

- Demography terms
- Reasons for growth
- Problems
- Control measures
- Surgical methods

Measuring a Population

Demography is the statistical study of human populations: their size, density, growth and distribution. A **census** is the official count of the population. In India it is held every ten years.

TERM	MEANING
Population density	Number of people living per unit area, usually per square kilometre
Birth rate	Number of live births per 1000 people in a year
Death rate	Number of deaths per 1000 people in a year
Growth rate	The rate at which a population changes: birth rate minus death rate (with migration, if counted). If births exceed deaths, the population grows.

$$\text{Growth rate} = \text{Birth rate} - \text{Death rate}$$



KEY POINT

Population explosion is a sudden, very rapid increase in the number of people, because the birth rate stays high while the death rate falls sharply.

- The world population passed **8 billion** in late 2022. India has over **1.4 billion** people and is now the most populous country, ahead of China.
- India holds about **17 to 18 per cent** of the world's people on only about **2.4 per cent** of its land area.
- The 2011 census counted about 1.21 billion people in India, a density of roughly 382 persons per square kilometre.

Why Did the Population Rise So Sharply?

REASON	EXPLANATION
Fall in death rate	Vaccines, antibiotics, hospitals, clean drinking water and sanitation have saved millions, especially infants. Epidemics such as smallpox and plague are controlled.
Better food supply	The Green Revolution and improved storage and transport reduced deaths from famine.
High birth rate	Early marriage, lack of education, especially of girls; poverty; a preference for sons; a belief that more children mean more earners; poor access to family planning.
Longer life expectancy	People live longer, so the number of people alive at one time rises.
Immigration	Entry of people from other countries or regions adds to a local population.

India's growth is helped by the large share of its population in the young, reproductive age group. Even if each couple has fewer children, the population keeps growing for a while. This effect is called **population momentum**.

Problems Caused by Population Explosion

- **Unemployment:** jobs do not grow as fast as the number of job seekers.
- **Over-exploitation of natural resources:** land, water, forests, minerals and fuels are used faster than they are replaced.
- **Low per capita income:** national income is shared among more people, lowering standard of living.
- **Rise in prices:** demand for food, housing and services rises faster than supply.
- **Pollution:** more vehicles, industries, waste and sewage.
- **Unequal distribution of wealth** and the spread of slums, poor health care and inadequate education.
- Shortage of food, housing, drinking water, schools and hospitals.



REAL LIFE

Rising population pushes people into crowded cities, and cities grow faster than their water supply, sewage and transport can cope with. Many large Indian cities already face summer water shortages.

Population Control

A stable population needs the birth rate to come down. The approach has several parts:

- **Education,** particularly of women, and raising the age of marriage (legal minimum 18 years for girls and 21 for boys).
- **Family planning:** promoting small families (the slogan 'Hum do, hamare do'), and providing information and services.
- **Better health care,** so that parents are confident their children will survive.
- **Incentives** by the government for small families, and public awareness through media.

Methods of birth control

TYPE	METHODS
Surgical (permanent)	**Vasectomy** in men and **tubectomy** in women
Barrier	Condoms, diaphragm
Intra-uterine device	Copper-T placed in the uterus
Hormonal	Oral contraceptive pills

	VASECTOMY	TUBECTOMY
Person	Male	Female
Operation	The two vasa deferentia are cut and tied (ligated) through a small incision in the scrotum	The two fallopian tubes (oviducts) are cut and tied through a small incision in the abdomen
Effect	Sperm cannot join the semen, so semen has no sperm	The egg cannot travel down the oviduct and sperm cannot reach it
Unchanged	Testosterone, sexual drive and ejaculation remain	Hormones, ovulation and menstrual cycle remain



BOARD ALERT

Vasectomy and tubectomy are **not** the same as castration or removal of the ovaries. Hormones are still made, and only the path of the gametes is blocked.



HOT QUESTION

India's birth rate has been falling for decades, yet the total population continues to increase. Give two reasons for this.



HOT QUESTION

Explain how an increase in population can lead to an increase in air and water pollution.

Mind Map Recap

BRANCH	WHAT TO REMEMBER
Terms	Demography, density, birth rate, death rate, growth rate
Reasons	Death rate falls, birth rate high, illiteracy, early marriage, better health
Problems	Unemployment, resources, low income, price rise, pollution, inequality
Control	Education, family planning, late marriage
Surgical	Vasectomy (vas deferens), tubectomy (oviducts)

Practice Questions

#	QUESTION	ANSWER
1	Define population density.	The number of persons living per unit area, such as per square kilometre.
2	What is population explosion?	A very rapid increase in population caused by a high birth rate along with a falling death rate.
3	Give two reasons for the decline of the death rate in India.	Better medical care and vaccination; improved sanitation and food supply.
4	Name two problems caused by overpopulation.	Unemployment and over-use of natural resources.
5	Differentiate between vasectomy and tubectomy.	Vasectomy cuts the vas deferens in males; tubectomy cuts the fallopian tubes in females.
6	How is growth rate calculated?	Growth rate = birth rate minus death rate.
7	How can the education of women help control population?	Educated women marry later, have fewer children and are more aware of family planning and health.
8	Why is a high population a problem for natural resources?	The resources are used faster than nature can replenish them.

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CHAPTER · BIOLOGY

Human Evolution

Humans did not appear suddenly. Fossils show a long line of ancestors who gradually stood upright, grew bigger brains and learned to make tools. This chapter covers the two classic theories of how life changes and the family tree that leads to us.



IN THIS CHAPTER

- Lamarck
- Darwin
- Peppered moth
- Human ancestors
- Vestigial organs

What Is Evolution?

KEY POINT

Organic evolution is the gradual change in living organisms over many generations, which leads to the formation of new species from earlier ones.

Evidence comes from fossils, from similar body structures in different animals, from vestigial organs, and from the similarity of DNA between related species. Two main theories try to explain how it happens.

Lamarck's Theory: Inheritance of Acquired Characters

Jean-Baptiste Lamarck, a French naturalist, put forward the first theory of evolution in 1809. It rests on two ideas.

- **Use and disuse of organs:** organs that are used a lot become better developed. Organs that are not used become smaller and may vanish.
- **Inheritance of acquired characters:** features gained by an individual during its life, because of use or disuse, pass to its offspring.

The giraffe example: Lamarck believed that the ancestors of the giraffe had short necks and forelimbs and fed on leaves on the ground. When the ground-level food ran out, they stretched to reach the high branches. The constant stretching made the neck and forelimbs a little longer, and the offspring inherited this extra length. After many generations the giraffe had a long neck.

Disuse: if an organ is not used, it shrinks. Examples in humans (see below): wisdom teeth, the vermiform appendix and the muscles that move the ear pinna.

Why Lamarck's theory is not accepted

- Changes acquired in the body (such as a bodybuilder's muscles) do not alter the DNA of the germ cells, so they are **not inherited**.
- August Weismann cut off the tails of mice for 22 generations and each new generation was born with tails of normal length.
- Variation arises from mutations and recombination, not from effort.

Vestigial Organs: Evidence of Disuse

Organs that have lost their original function and have become reduced are called **vestigial organs**.

- **Wisdom teeth:** the third molars, reduced because our ancestors' diet of tough raw food was replaced by softer food.
- **Vermiform appendix:** a small remnant of the large caecum of herbivorous ancestors, in which cellulose is digested.
- **Pinnae (external ear muscles):** most humans cannot move their ears, unlike many animals that turn the pinna towards a sound.

KEY POINT

Use of organs leads to development (the neck and forelimbs of a giraffe). Disuse leads to reduction (vestigial organs such as the wisdom tooth, appendix and pinna).

Darwin's Theory of Natural Selection

Charles Darwin, after his voyage on HMS *Beagle* (1831 to 1836), published *On the Origin of Species* in 1859. Alfred Russel Wallace reached the same idea independently.

1. Overproduction
Organisms produce many more offspring than can survive.

2. Struggle for existence
Limited food, space and mates mean that they compete with each other, and with predators and disease.

3. Variation
No two individuals are exactly alike. Some variations help survival in the environment.

4. Survival of the fittest
Individuals with useful variations are better adapted and more likely to survive and breed. The others die before they leave offspring.

5. Inheritance
The successful variations are passed to the next generation. Over many generations they build up and give rise to new species.

Giraffe by Darwin: some ancestral giraffes happened to have slightly longer necks. When food on the lower branches became scarce, they could reach the leaves and survived and reproduced. Their offspring inherited the longer necks. The short-necked ones starved. Repeated over generations, the average neck got longer.

Example: the peppered moth (*Biston betularia*)

STAGE	WHAT HAPPENED
Before the Industrial Revolution (England)	Most peppered moths were **light coloured**, which camouflaged them against lichen-covered tree trunks. Dark moths were rare, because birds spotted and ate them easily.
During industrialisation	Factory soot killed the lichen and blackened the tree trunks. Now **dark moths** were camouflaged and light moths were eaten by birds. Dark moths became the majority.
After pollution control	As the air became cleaner, lichens returned and light moths increased again.

KEY POINT

The change in the moth population shows **natural selection**. The environment 'selected' the form that was best camouflaged. The moths did not change colour by themselves: the useful variation (dark colour) already existed, and it became common because it helped survival.

	LAMARCK	DARWIN
Cause of change	Effort, use and disuse of organs	Natural selection acting on variations
Inheritance of acquired traits	Yes	No; only inherited variations count
Role of variation	Not discussed	Central
Giraffe's neck	Stretching lengthened the neck, and this was inherited	Naturally longer-necked giraffes survived better and left more offspring
Status today	Rejected	Accepted (with modern genetics)



SIDEKICK TIP

Lamarck = 'Use it or lose it', Darwin = 'Fit survive, then pass it on'. The key word for Darwin is **selection**.

Human Ancestors

Fossils show that the ancestors of modern humans and the apes shared a common ancestor several million years ago. Humans did **not** descend from the chimpanzee or from any ape living today.

ANCESTOR	TIME (APPROX.)	CRANIAL CAPACITY	KEY FEATURES
Australopithecus	About 4 to 2 million years ago, Africa	About 400 to 500 cc	Walked upright on two legs (**bipedal**) but had a small, ape-like skull, large jaws and heavy brow ridge. Small canines compared to apes. Height about 1.1 to 1.4 m. 'Lucy' is a famous fossil.
Homo habilis ('handy man')	About 2.4 to 1.5 million years ago	About 600 to 700 cc	Walked upright; first to make and use simple stone tools; smaller teeth and jaws
Homo erectus ('upright man')	About 1.9 million to 100,000 years ago	About 800 to 1100 cc	Fully erect posture, tall (up to 1.6 to 1.8 m), used fire, made better tools, spread out of Africa into Asia and Europe. Heavy brow ridge, no chin.
Neanderthal man	About 400,000 to 40,000 years ago, Europe and West Asia	About 1200 to 1700 cc (slightly larger than ours)	Stocky and strong; heavy brow ridges; sloping forehead; no true chin; wore clothing, made tools, buried the dead
Cro-Magnon man	About 40,000 to 10,000 years ago, Europe	About 1400 to 1600 cc	Anatomically modern: tall (about 1.8 m), straight forehead, a well-formed chin, small brow ridges, and little body hair. Made cave paintings, advanced tools
Homo sapiens sapiens (modern man)	About 300,000 years ago, in Africa, to the present	About 1350 to 1400 cc	Erect posture, high rounded forehead, no or very light brow ridge, a distinct chin, small canines and teeth, little body hair, language, art, agriculture and technology

Trends in human evolution

FEATURE	CHANGE OVER TIME
Bipedalism	From knuckle-walking or tree climbing to fully upright walking on two legs, freeing the hands
Cranial capacity	Increased from about 400 cc to about 1350 cc, so more brain power
Canine teeth	Became smaller and flatter as the diet changed and tools replaced teeth as weapons
Forehead and brow ridges	The forehead became high and rounded; the heavy brow ridges disappeared
Chin	Absent in early forms; well-developed in modern humans
Body hair	Reduced
Height and posture	Increased in height; posture became fully erect with a straight spine

FUN FACT

The human brain is only about 2 per cent of body weight but uses roughly a fifth of the body's energy.

BOARD ALERT

Do not write that 'man evolved from monkeys'. Say that humans and apes have a **common ancestor**.

HOT QUESTION

Industrial pollution is cleaned up in a town. Predict how the peppered moth population in that town will change over many years, and explain using Darwin's idea.

HOT QUESTION

A bodybuilder develops big muscles by training for years. Will his children be born with larger muscles? Which theory says yes, and why is it wrong?

Mind Map Recap

BRANCH	WHAT TO REMEMBER
Lamarck	Use and disuse + inheritance of acquired traits; giraffe; rejected
Darwin	Overproduction, struggle, variation, survival of the fittest, inheritance
Peppered moth	Light vs dark; soot; camouflage; natural selection
Ancestors	Australopithecus → H. habilis → H. erectus → Neanderthal / Cro-Magnon → H. sapiens sapiens
Trends	Bipedal, bigger brain, smaller canines, chin, less hair
Vestigial	Wisdom teeth, appendix, pinnae

Practice Questions

#	QUESTION	ANSWER
1	State Lamarck's theory in two lines.	Organs used regularly become well developed and those not used degenerate; such acquired changes are inherited by the offspring.
2	Why is Lamarck's theory rejected?	Acquired characters do not change the DNA of gametes, so they are not inherited.
3	What is natural selection?	The process by which individuals with favourable variations survive and reproduce more, so that those variations become common.
4	Explain the evolution of the long neck of the giraffe according to Darwin.	Giraffes with slightly longer necks reached food and survived; they left offspring with long necks, and so the average neck length rose over generations.
5	Why did dark peppered moths become common during industrialisation?	Soot darkened the tree trunks, so dark moths were camouflaged and escaped predators, while light moths were eaten.
6	Name the human ancestor that first made stone tools.	Homo habilis.
7	Which human ancestor first used fire and spread out of Africa?	Homo erectus.
8	What is a vestigial organ? Give two examples.	A reduced organ that has lost its original function, e.g. the vermiform appendix and wisdom teeth.
9	State three trends seen in the evolution of humans.	Walking upright, increasing cranial capacity, reduction in canine size (also: chin, loss of body hair).

15

CHAPTER · BIOLOGY

Pollution

Air, water and soil can clean themselves only up to a limit. When we add more than they can absorb, we call it pollution. This chapter covers the types, sources and effects of pollution, and the measures that help.



IN THIS CHAPTER

- Air, water, soil
- Thermal, radiation, noise
- Biomedical waste
- Greenhouse and ozone
- Acid rain
- Control measures

What Is Pollution?

KEY POINT

Pollution is the undesirable change in the physical, chemical or biological characteristics of air, water or soil that harms living things. A substance that causes pollution is a **pollutant**.

Air Pollution

SOURCE	MAIN POLLUTANTS	EFFECTS
Vehicles	Carbon monoxide, oxides of nitrogen, unburnt hydrocarbons, lead (from leaded petrol), particles	CO binds to haemoglobin and reduces oxygen supply; smog; breathing problems
Industries	Sulphur dioxide, oxides of nitrogen, smoke, fly ash, chemicals	Acid rain; respiratory diseases; damage to plants
Burning garbage	Smoke, fine particles, carbon monoxide, toxic gases	Eye and lung irritation; poisonous gases from burning plastics
Brick kilns	Suspended particles, sulphur dioxide, carbon dioxide, soot	Particles cause asthma and lung disease; soot deposits on leaves and soil

Water Pollution

SOURCE	EFFECTS
Household detergents	Phosphates in detergents over-fertilise water (eutrophication): algae multiply, then die and decay, using up the oxygen. Fish die. Detergents are often slow to break down.
Sewage	Contains germs that cause cholera, typhoid and dysentery. Decay of sewage uses up oxygen and releases foul smells.
Industrial waste	Heavy metals (mercury, lead), acids and chemicals. These poison aquatic life and enter food chains.
Oil spills	Oil forms a film on water that cuts off light and oxygen and coats the feathers of birds and the gills of fish.

REAL LIFE

Mercury discharged by a factory in Minamata Bay, Japan, built up in fish and caused a serious nerve disease in the people who ate them. The incident is a standard example of how pollutants travel through the food chain.

Thermal Pollution

Power plants and factories take in water from rivers or lakes to cool their machines and then return it warm. Warmer water holds **less dissolved oxygen**, and it also speeds up the metabolism of aquatic animals, so they need more oxygen when it is scarcer. Fish and other organisms may die. It is prevented by cooling towers or cooling ponds before the water is returned.

Soil Pollution

- **Industrial waste:** toxic chemicals and metals in sludge and ash.
- **Urban, commercial and domestic waste:** plastics, glass, metal, packaging, kitchen waste dumped in landfills.
- **Chemical fertilisers and pesticides:** excess fertiliser changes the soil chemistry, runs off into water, and kills useful soil organisms such as earthworms. Pesticides such as **DDT** do not break down quickly. They build up as they pass along the food chain, a process called **biomagnification**, and thin the eggshells of birds of prey.

Biomedical Waste

Waste produced in hospitals and clinics: used and discarded needles, syringes, soiled dressings, bandages, blood, body parts and expired medicines. It can spread diseases such as hepatitis B and HIV. It must be collected in colour-coded bags or containers, disinfected, and then safely destroyed (by incineration or autoclaving), and not mixed with ordinary garbage.

Radiation Pollution

SOURCE	EFFECT
X-rays (medical use)	Over-exposure can damage cells and cause burns, cancer and mutations. Health workers use lead aprons.
Radioactive fallout from nuclear plants, accidents and weapons tests (e.g. Chernobyl 1986, Fukushima 2011)	Radioactive elements contaminate air, water, soil and crops. Cause cancer, birth defects and genetic mutations that can last for generations.

Noise Pollution

Unwanted loud sound is measured in **decibels (dB)**. Sources include motor vehicles and horns, industrial machinery, construction sites and loudspeakers at festivals. Sustained exposure above about 85 to 90 dB can damage hearing. Other effects: headache, stress, high blood pressure, loss of sleep and poor concentration. Control: silencers, no-horn zones, restrictions on loudspeakers, sound barriers and planting trees.

Biodegradable and Non-biodegradable Waste

	BIODEGRADABLE	NON-BIODEGRADABLE
Meaning	Broken down by microorganisms (bacteria, fungi) into harmless substances	Not broken down by microorganisms, or only extremely slowly
Examples	Paper, vegetable and fruit peels, cotton, wood, animal waste, leftover food	Plastics, glass, Styrofoam, metal cans, DDT and many pesticides, aluminium foil
Management	Composting, biogas production	Reduce, reuse, recycle

SIDEKICK TIP

Segregate at source: a green bin for wet (biodegradable) waste and a blue bin for dry (recyclable) waste. It makes composting and recycling possible.

Effects of Pollution on Climate and Environment

1. Greenhouse effect and global warming

Certain gases in the air let sunlight through but trap the heat that the Earth radiates. This warming is natural, and it keeps the Earth warm enough for life. Human activities have added to the gases, mainly **carbon dioxide** (burning of fossil fuels, deforestation), methane (paddy fields, cattle, landfills), nitrous oxide and **CFCs**. The extra trapped heat causes **global warming**.

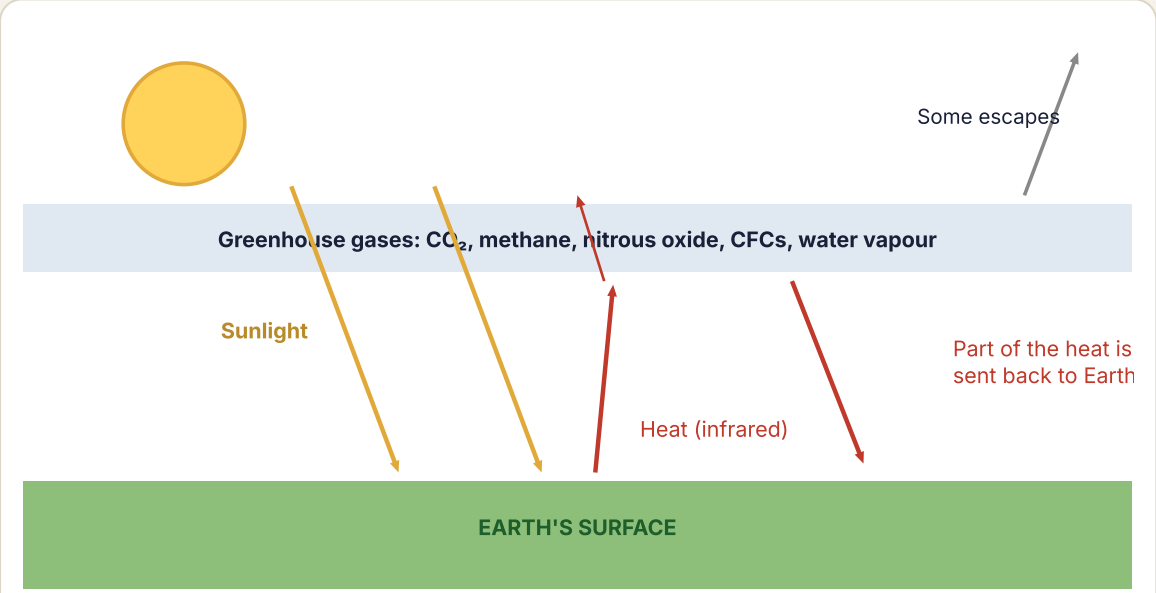


FIG 15.1 THE GREENHOUSE EFFECT

- **Effects of global warming:** melting of glaciers and polar ice, rising sea levels, flooding of coastal areas, more frequent heat waves, droughts and storms, changes in crop patterns, and loss of species.

2. Acid rain

Sulphur dioxide and oxides of nitrogen from power plants, factories and vehicles dissolve in rain water to form sulphuric acid and nitric acid. Rain with pH below about 5.6 is acid rain.

- It damages buildings and monuments made of marble and limestone. The Taj Mahal is affected ('marble cancer').
- It acidifies lakes and kills fish; harms forests; makes soil less fertile by washing out nutrients; corrodes metal structures.

3. Ozone layer depletion

The ozone layer in the stratosphere (about 15 to 35 km above the Earth) absorbs most of the harmful **ultraviolet (UV-B)** rays of the sun. **Chlorofluorocarbons (CFCs)**, once used in refrigerators, air conditioners, aerosol sprays and foam, release chlorine atoms in the stratosphere, and a single chlorine atom can destroy thousands of ozone molecules. This has created the 'ozone hole' over Antarctica.

- **Effects:** more UV reaches the Earth, causing skin cancer, cataract, weakened immunity, damage to crops and to plankton in the sea.
- The **Montreal Protocol** (1987) is an international agreement to phase out CFCs.

BOARD ALERT

Do not mix up the two atmospheric problems. **Greenhouse gases** (CO_2, CH_4) cause global warming. **CFCs** cause ozone depletion. Ozone depletion is not the cause of global warming.

Measures to Control Pollution

MEASURE	HOW IT HELPS
Unleaded petrol and CNG	Lead is removed from vehicle exhaust. CNG burns cleanly and releases much less carbon monoxide, particles and hydrocarbons
Switching off engines at traffic lights	Cuts fuel use and exhaust fumes
Social forestry	Growing trees on community and waste land; trees absorb CO_2 and dust, prevent erosion and give fuel wood and fodder
Sewage treatment plants	Sewage is screened and settled (primary), digested by bacteria (secondary), and disinfected (tertiary) before release into rivers
Ban on polythene and plastics	Plastic is non-biodegradable, blocks drains and harms animals that eat it
Organic farming	Use of compost, manure and natural pest control instead of chemical fertilisers and pesticides
Euro / Bharat standards	Rules that set limits for the pollutants in vehicle exhaust. India has moved to **Bharat Stage VI** norms, comparable to Euro 6

Swachh Bharat Abhiyan

Swachh Bharat Abhiyan (Clean India Mission) was launched by the Government of India on **2 October 2014**, the birth anniversary of Mahatma Gandhi. It aims to end open defecation, build toilets, manage solid waste and keep villages, towns and cities clean, with the participation of every citizen.

FUN FACT

A plastic bottle may take hundreds of years to break down. Glass can take much longer, but both can be recycled again and again.

HOT QUESTION

A lake near a town develops a thick green layer of algae in summer, and then many fish die. Trace the sequence of events that connects detergent-rich sewage with the death of fish.

HOT QUESTION

A student says: 'Banning CFCs will stop global warming.' Evaluate this statement.

Mind Map Recap

BRANCH	WHAT TO REMEMBER
Air	Vehicles, industry, garbage, brick kilns: CO, SO2, NOx, particles
Water	Detergents, sewage, industrial waste, oil spills
Other types	Thermal, soil, biomedical, radiation, noise
Waste	Biodegradable vs non-biodegradable; DDT biomagnification
Atmosphere	Greenhouse effect → global warming; acid rain; ozone depletion (CFCs)
Control	CNG, unleaded petrol, forests, STPs, plastic ban, organic farming, Bharat VI, Swachh Bharat

Practice Questions

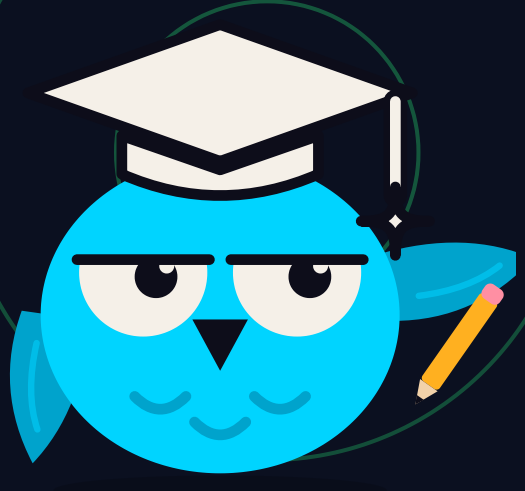
#	QUESTION	ANSWER
1	Define pollutant with two examples.	A substance that causes pollution, e.g. carbon monoxide in air and oil in water.
2	Why does thermal pollution kill fish?	Warm water holds less oxygen and raises the fishes' oxygen need.
3	What is meant by biodegradable? Give two examples.	Able to be broken down by microbes; e.g. paper and vegetable peels.
4	Name two greenhouse gases and one source of each.	Carbon dioxide from burning fuel; methane from paddy fields or cattle.
5	What causes acid rain?	Dissolution of sulphur dioxide and nitrogen oxides in rain water, forming sulphuric and nitric acids.
6	Which chemicals deplete the ozone layer, and what is the harm?	CFCs; more UV reaches the Earth, causing skin cancer and cataract.
7	Why is CNG a better fuel than petrol?	It burns more completely, with less CO and fewer particles and hydrocarbons.
8	What is biomagnification? Give an example.	Increase in the concentration of a non-degradable substance along the food chain, e.g. DDT.
9	Mention two ways to reduce noise pollution.	Banning use of loudspeakers at night; using silencers on vehicles; planting trees.
10	What is the aim of Swachh Bharat Abhiyan?	To make India clean by ending open defecation and managing waste.

16

CHAPTER · BIOLOGY

Practical Lab Companion

Twenty marks of Internal Assessment are won in the lab, not in the exam hall. Here is every experiment and identification task from the 2027 syllabus, with the method, the observation and the conclusion you should write.



IN THIS CHAPTER

- Mitosis slides
- Diffusion and osmosis
- Transpiration
- Photosynthesis
- Identification tasks
- Grading criteria

Plant Life: Experiments You Must Be Able to Perform

TOPIC	EXPERIMENT
Cell division	Observation of permanent slides of stages of mitosis
Diffusion	Using potassium permanganate in water
Osmosis	Thistle funnel experiment and potato osmoscope
Absorption	Using a small herbaceous plant
Transpiration	Bell jar; unequal transpiration in a dorsiventral leaf with cobalt chloride paper; Ganong's potometer
Photosynthesis	Necessity of light, carbon dioxide and chlorophyll; release of oxygen using Hydrilla or Elodea

Mitosis Slides

Use a permanent slide of an **onion root tip**. Focus first under low power and find the region just behind the root cap, where cells are small, square and dividing. Switch to high power and identify the stage by the position of the chromosomes.

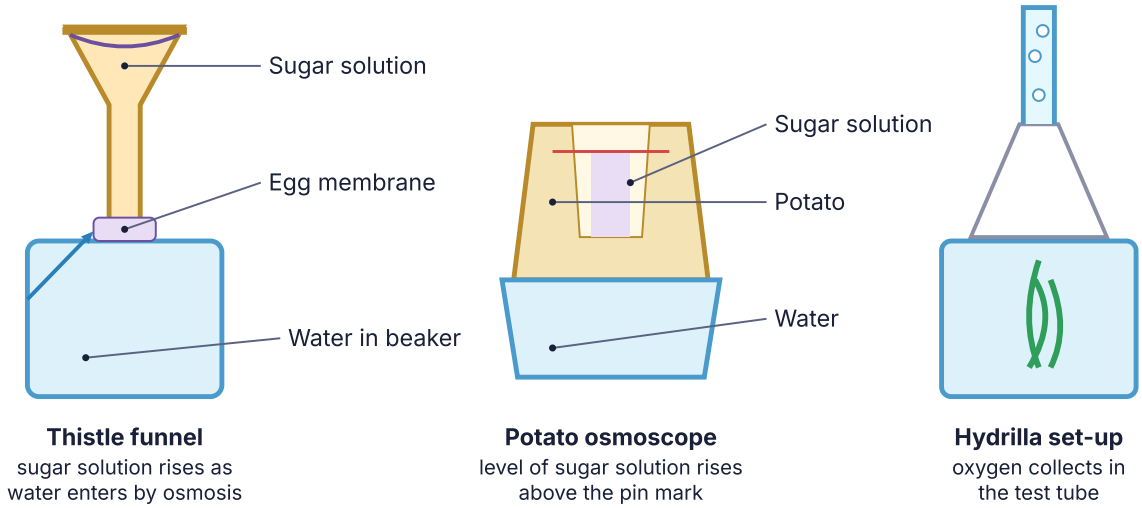
STAGE	WHAT YOU SEE
Prophase	Thread-like, coiled chromosomes; nucleolus and nuclear membrane fading
Metaphase	Chromosomes in a line across the middle of the cell
Anaphase	V-shaped chromatids moving toward opposite poles
Telophase	Two groups of chromosomes at the poles; a cell plate forming between them



SIDEKICK TIP

Count only **one** clear cell and label it. A drawing of a single, well-chosen cell scores better than a crowded sketch of many.

Diffusion, Osmosis and Absorption



Thistle funnel
sugar solution rises as water enters by osmosis

Potato osmoscope
level of sugar solution rises above the pin mark

Hydrilla set-up
oxygen collects in the test tube

FIG 16.1 THREE CLASSIC LAB SET-UPS (SCHEMATIC)

EXPERIMENT	PROCEDURE	OBSERVATION AND CONCLUSION
Diffusion ($KMnO_4$)	Drop a crystal into a beaker of still water; do not stir	Purple colour spreads from the crystal outward until uniform. Molecules move from high to low concentration: diffusion.
Thistle funnel	Tie egg membrane over the wide mouth of the funnel, fill with sugar solution and dip in water; note the level	The level rises in the tube. Water has entered through the semi-permeable membrane by osmosis.
Potato osmoscope	Cut a cavity in a peeled potato, fill with sugar solution, set in water and mark the level with a pin	Level rises: osmosis. A boiled potato (control) shows no rise.
Absorption (herbaceous plant)	Place a small plant with its roots in a beaker of water coloured with red ink or eosin, in sunlight	After some time, the stem's xylem is stained red. Roots absorb water and it moves up the xylem.

Transpiration

Bell jar experiment

1. Set up
Water a potted plant well. Cover the soil and pot with a polythene sheet so that evaporation from the soil is blocked.

2. Cover
Place a clean, dry bell jar over the plant and seal its base with Vaseline on a glass plate.

3. Control
Set up an identical bell jar with only the pot, no plant.

4. Observe
After a few hours, droplets of water appear on the inner walls of the jar with the plant. The control stays dry.



KEY POINT

Water droplets form only in the jar containing the plant. This proves that the water vapour came from the plant, through transpiration, and not from the soil or the pot.



BOARD ALERT

Always cover the soil and the pot. If you do not, the examiner can say that the water came from evaporation of the soil.

Cobalt chloride paper

Dry blue paper on both surfaces of a leaf, covered by glass slides. Paper on the lower surface turns pink first: more stomata on the lower surface. Keep the paper dry with forceps before use (blue colour), and compare the time taken.

Ganong's potometer

Fill the apparatus with water and no air bubbles, cut the shoot under water and insert it with a tight cork. Introduce one air bubble into the capillary. Note the distance moved by the bubble in a fixed time (e.g. 5 minutes), first in room conditions, and then in conditions such as fan, sunlight or shade. Reset the bubble with the reservoir tap each time. Result: the rate of uptake (and so of transpiration) is highest in strong light with a fan, and lowest in a humid shade.

Photosynthesis

See the chapter on photosynthesis for the full table. Remember the sequence for the starch test.

1. Destarch
Keep the potted plant in complete darkness for 24 to 48 hours.

2. Expose
Set up the condition being tested and a control, then expose to light for some hours.

3. Kill
Dip the leaf in boiling water for about a minute.

4. Decolourise
Boil the leaf in methylated spirit in a water bath until the green colour has gone.

5. Wash
Rinse the leaf in warm water to soften it.

6. Test
Spread the leaf in a dish and add iodine solution. A blue-black colour shows that starch is present.



BOARD ALERT

Never heat methylated spirit directly on a flame. Always use a water bath, because the spirit is highly inflammable.

EXPERIMENT	SET-UP	RESULT	CONCLUSION
Light	Part of a destarched leaf covered with black paper	Covered part brown; uncovered part blue-black	Light is necessary
CO_2	Destarched plant in a bell jar with KOH; control with water	Leaf in KOH stays brown; control blue-black	CO_2 is necessary
Chlorophyll	Variegated leaf	Only the green regions turn blue-black	Chlorophyll is necessary
Release of O_2	Hydrilla under an inverted funnel and test tube, in sunlight	Bubbles of gas collect; a glowing splinter relights	O_2 is released

Animal Life: Identification Tasks

- Identify structures of the **urinary system, heart and kidney (internal structure)** and the **brain (external view)** using models and charts.
- Identify the different **types of blood cells** under a microscope: RBCs are the most numerous, round and without a nucleus; WBCs have a stained nucleus and are larger; platelets are tiny fragments.
- Identify the internal structure of the **ear and eye** using models and charts.
- Identify and locate the **adrenal, pancreas, thyroid and pituitary** glands using a model or chart.



SIDEKICK TIP

For a spotting question, write the name, one feature that helped identify it, and one function. A complete answer is worth the marks even if the diagram is small.

How the 20 Marks Are Awarded

EXAMINER	MARKS
Subject teacher (Internal Examiner)	10
External Examiner	10
Total sent to CISCE	20



KEY POINT

The internal and external examiners assess the practical work and project work independently. The school then sends the total out of 20 to CISCE.

The Five Grading Criteria

CRITERION	WHAT THE EXAMINER LOOKS FOR
Preparation	Follows instructions with understanding and uses apparatus safely and confidently
Procedure / Testing	Plans the investigation logically and controls the variables
Observation	Records data systematically and accurately
Inference / Results	Draws correct conclusions and recognises sources of error
Presentation	Neat, accurate, well-labelled diagrams and records



SIDEKICK TIP

Label diagrams in pencil, draw label lines with a ruler, and always write the heading and magnification. Presentation is one of the five criteria, so neatness earns marks.

Mind Map Recap

BRANCH	WHAT TO REMEMBER
Mitosis slides	Onion root tip; identify four stages
Osmosis	Thistle funnel, potato osmoscope (boiled control)
Transpiration	Bell jar, cobalt chloride, potometer
Photosynthesis	Destarch; light, CO_2, chlorophyll; Hydrilla O_2
Identification	Urinary system, heart, kidney, brain, blood cells, eye, ear, glands

Practice Questions

#	QUESTION	ANSWER
1	What is the control in the bell jar experiment?	A bell jar with only the pot and soil, no plant.
2	What colour is cobalt chloride paper when dry and when moist?	Blue when dry; pink when moist.
3	Why is the potato osmoscope repeated with a boiled potato?	Boiling kills the cells, so no osmosis occurs; it is a control.
4	What does an air bubble in a potometer show?	The movement of the bubble shows the rate of water uptake.
5	Why is the plant destarched before an experiment on photosynthesis?	So that any starch found at the end was made during the experiment.
6	Which chemical is used to absorb CO_2 in the experiment?	Potassium hydroxide (KOH).
7	Why is a variegated leaf used?	To show that only the green (chlorophyll-containing) parts make starch.
8	State the test for starch and its positive result.	Add iodine solution; blue-black colour.

Done reading?
Now go break the board.



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